# Grover's Algorithm

Assessment of quantum search

**ANSWER BOOK**

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key** with the topic tested by every question (page 2), and
2. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution: concepts required, relevant formulae, step-by-step explanation, why each other option is incorrect, and a key takeaway (from page 4).

## Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

| | |
|---|---|
| *Full marks*: | +3 if ONLY the correct option is chosen; |
| *Zero marks*: | 0 if none of the options is chosen (question unanswered); |
| *Negative marks*: | −1 in all other cases (wrong option, or more than one option chosen). |

## How to evaluate

For each section count $C$ = number of correct answers and $W$ = number of questions answered wrongly (or with more than one bubble darkened). Section marks = $3C - W$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (NumPy, including gate-by-gate simulation of the Grover circuits, and CUDA-Q running the notebook's own kernels on the CPU simulator) by `verify_answers.py` in the source folder.

## Conventions (identical to the Question Paper)

- $n$ qubits, $N = 2^n$ items, $M$ marked items ($M = 1$ unless stated otherwise). A basis state is written $|x\rangle = |x_1 x_2 \ldots x_n\rangle$: **qubit 1 is the most significant bit** and is written first. (The accompanying notebook writes `N_QUBITS` for $n$ and $M = 2^N$ for the number of items, and $O$ for the oracle.)
- $|s\rangle = H^{\otimes n}|0\ldots0\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle$. The oracle is $U_\omega|x\rangle = (-1)^{f(x)}|x\rangle$ with $f(x) = 1$ exactly on the marked items; the diffusion is $D = 2|s\rangle\langle s| - I$; one iteration is $G = DU_\omega$ (oracle first).
- $\sin\theta = \sqrt{M/N}$ with $0 < \theta \le \frac{\pi}{2}$; $|m\rangle$ and $|r\rangle$ are the uniform superpositions of the marked and the unmarked items. Then $G^k|s\rangle = \sin((2k+1)\theta)|m\rangle + \cos((2k+1)\theta)|r\rangle$, so $P_k = \sin^2((2k+1)\theta)$ and $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.
- **Textbook circuit:** $H^{\otimes n}$, then $k$ times: [$X$ on each qubit where $x^*$ has a 0; a multi-controlled $Z$ on all $n$ qubits; the same $X$ gates] (no ancillas) followed by [$H^{\otimes n}$; $X^{\otimes n}$; a multi-controlled $Z$; $X^{\otimes n}$; $H^{\otimes n}$]; finally measure all qubits.
- $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$; $X, Y, Z$ are the Pauli matrices; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- All measurements are in the computational basis unless stated otherwise; log means $\log_2$. A numerical value given in a question may be used without a calculator.

# Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

## Section A — Mathematics and Search Foundations

| Q. | Ans. | Topic tested | Q. | Ans. | Topic tested |
|---|---|---|---|---|---|
| 1 | **B** | Triple-angle identity | 11 | **A** | The signed sum $N - 2M$ |
| 2 | **C** | Reflection matrix $2uu^T - I$ | 12 | **B** | Normalising a state vector |
| 3 | **D** | Two reflections make a rotation | 13 | **B** | Square-root scaling |
| 4 | **D** | Eigenvalues of a rotation | 14 | **C** | Complex inner product |
| 5 | **A** | Powers of a complex phase | 15 | **C** | $\sin^2 3\theta$ as a polynomial |
| 6 | **D** | Optimal integer $k$; why $-\frac{1}{2}$ matters | 16 | **D** | Traces of projectors and reflections |
| 7 | **A** | Inversion about the mean | 17 | **C** | Rounding rule vs. $\arcsin x > x$ |
| 8 | **C** | Classical search without replacement | 18 | **A** | Spectrum of a reflection |
| 9 | **A** | Binary vs. linear search | 19 | **D** | First maximum and period of $p(k)$ |
| 10 | **B** | Counting solutions of a Boolean function | 20 | **B** | Tolerance window around the peak |

## Section B — Quantum Computing Foundations

| Q. | Ans. | Topic tested | Q. | Ans. | Topic tested |
|---|---|---|---|---|---|
| 21 | **B** | Uniform state, partial outcome | 31 | **A** | Overlap $\cos\theta$ |
| 22 | **A** | Tensor-product amplitude | 32 | **D** | State-vector memory |
| 23 | **C** | Hadamard layer signs | 33 | **A** | Diffusion for one qubit |
| 24 | **C** | Phase kickback with $\|-\rangle$ | 34 | **D** | $\|s\rangle$ is fixed by $D$ |
| 25 | **D** | No kickback with $\|+\rangle$ | 35 | **C** | Single-qubit gate actions |
| 26 | **C** | $X$ conjugation selects the item | 36 | **B** | Product vs. entangled states |
| 27 | **D** | CCZ from Toffoli | 37 | **A** | Why the oracle alone reveals nothing |
| 28 | **A** | Global vs. relative phase | 38 | **C** | Relative phases are observable |
| 29 | **B** | Subspace measurement | 39 | **D** | Toffoli as a phase oracle |
| 30 | **B** | Unitary vs. projector | 40 | **B** | Ancilla stays unentangled |

## Section C — Grover's Algorithm Fundamentals

| Q. | Ans. | Topic tested | Q. | Ans. | Topic tested |
|---|---|---|---|---|---|
| 41 | **D** | $N = 4$ is exact in one step | 51 | **D** | $M = 3N/4$ |
| 42 | **B** | One iteration, $N = 8$ | 52 | **A** | Over-rotation at $2k_{\text{opt}}$ |
| 43 | **D** | $k_{\text{opt}}$ and $P$ for $n = 4$ | 53 | **A** | Size of the speed-up |
| 44 | **B** | Matrix of $D$ for $N = 4$ | 54 | **B** | Shot statistics |
| 45 | **C** | Building the oracle | 55 | **A** | $k_{\text{opt}}$ against $n$ |
| 46 | **A** | Gate count per iteration | 56 | **A** | Blocks of the circuit |
| 47 | **B** | Diffusion circuit $= -D$ | 57 | **B** | Independence of the marked item |
| 48 | **C** | Unmarked amplitude | 58 | **D** | Oscillation and $\sqrt{N}$ scaling |
| 49 | **C** | $M = N/4$ | 59 | **C** | Amplitude trace, first iteration |
| 50 | **D** | $M = N/2$ | 60 | **C** | Amplitude trace, second iteration |

## Section D — Advanced Grover and Implementation

| Q. | Ans. | Topic tested | Q. | Ans. | Topic tested |
|---|---|---|---|---|---|
| 61 | **B** | Several solutions: $k_{\text{opt}}$ | 71 | **B** | Simulation cost |
| 62 | **A** | BBHT threshold for unknown $M$ | 72 | **C** | Reading the sweep code |
| 63 | **D** | Exact Grover search | 73 | **D** | Interpreting 2000/2000 shots |
| 64 | **D** | Optimality (BBBV, Zalka) | 74 | **C** | Repeat-until-success cost |
| 65 | **C** | Reading gate statistics | 75 | **A** | Query complexities |
| 66 | **B** | Amplitude amplification cost | 76 | **B** | Variants of Grover |
| 67 | **D** | Notebook memory estimate | 77 | **D** | Grover and NP-completeness |
| 68 | **A** | Key search | 78 | **A** | Unknown $M$; flat curve at $M = N/2$ |
| 69 | **C** | Noise and depth | 79 | **B** | Quantum counting outcomes |
| 70 | **A** | Toffoli count | 80 | **C** | Quantum counting estimate |

## Quick-marking grid

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 **B** | 2 **C** | 3 **D** | 4 **D** | 5 **A** | 6 **D** | 7 **A** | 8 **C** | 9 **A** | 10 **B** |
| 11 **A** | 12 **B** | 13 **B** | 14 **C** | 15 **C** | 16 **D** | 17 **C** | 18 **A** | 19 **D** | 20 **B** |
| 21 **B** | 22 **A** | 23 **C** | 24 **C** | 25 **D** | 26 **C** | 27 **D** | 28 **A** | 29 **B** | 30 **B** |
| 31 **A** | 32 **D** | 33 **A** | 34 **D** | 35 **C** | 36 **B** | 37 **A** | 38 **C** | 39 **D** | 40 **B** |
| 41 **D** | 42 **B** | 43 **D** | 44 **B** | 45 **C** | 46 **A** | 47 **B** | 48 **C** | 49 **C** | 50 **D** |
| 51 **D** | 52 **A** | 53 **A** | 54 **B** | 55 **A** | 56 **A** | 57 **B** | 58 **D** | 59 **C** | 60 **C** |
| 61 **B** | 62 **A** | 63 **D** | 64 **D** | 65 **C** | 66 **B** | 67 **D** | 68 **A** | 69 **C** | 70 **A** |
| 71 **B** | 72 **C** | 73 **D** | 74 **C** | 75 **A** | 76 **B** | 77 **D** | 78 **A** | 79 **B** | 80 **C** |

# Answer Bank — questions with detailed solutions

## SECTION A · Mathematics and Search Foundations Q.1–20

**Q.1** SINGLE-CORRECT MCQ · Triple-angle identity

If $\sin\theta = \frac{1}{4}$, the value of $\sin^2 3\theta$ is

**(A)** $\frac{1}{16}$ **(B)** ✓ $\frac{121}{256}$ **(C)** $\frac{9}{16}$ **(D)** $\frac{11}{16}$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Triple-angle identity $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. In Grover's algorithm $P_1 = \sin^2 3\theta$.

**Relevant formulae.** $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, where $\theta$ is any angle.

**Step-by-step explanation.** Use the triple-angle identity $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$:

$$\sin 3\theta = \frac{3}{4} - 4\cdot\frac{1}{64} = \frac{12}{16} - \frac{1}{16} = \frac{11}{16}, \qquad \sin^2 3\theta = \frac{121}{256}.$$

This is exactly the success probability of one Grover iteration on $N = 16$ items. The value $\frac{9}{16} = (3\sin\theta)^2$ is the small-angle estimate; it ignores the cubic correction.

**Why the other options are incorrect.**

**(A)** $\frac{1}{16} = \sin^2\theta$ is the probability before any iteration.

**(C)** $\frac{9}{16} = (3\sin\theta)^2$ uses the small-angle estimate $\sin 3\theta \approx 3\sin\theta$ and drops $-4\sin^3\theta$.

**(D)** $\frac{11}{16}$ is $\sin 3\theta$ itself: the final squaring was forgotten.

**Key takeaway.** Keep the exact identity for small $N$; the small-angle estimate is only reliable for large $N$.

**Q.2** SINGLE-CORRECT MCQ · Reflection matrix $2uu^T - I$

Let $u = (\frac{3}{5}, \frac{4}{5})^T$. The matrix $2uu^T - I$ equals

**(A)** $\frac{1}{25}\begin{pmatrix}18 & 24\\ 24 & 32\end{pmatrix}$ **(B)** $\frac{1}{25}\begin{pmatrix}-7 & 12\\ 12 & 7\end{pmatrix}$ **(C)** ✓ $\frac{1}{25}\begin{pmatrix}-7 & 24\\ 24 & 7\end{pmatrix}$ **(D)** $\frac{1}{25}\begin{pmatrix}7 & -24\\ -24 & -7\end{pmatrix}$

**Correct answer: (C)** **Detailed solution**

**Concepts required.** $uu^T$ is the matrix with entries $u_i u_j$. $2uu^T - I$ reflects vectors in the line along $u$: it fixes $u$ and negates the perpendicular direction.

**Relevant formulae.** $(uu^T)_{jk} = u_j u_k$ for a real unit vector $u$; $I$ is the identity matrix. $R = 2uu^T - I$ satisfies $Ru = u$.

**Step-by-step explanation.** $uu^T = \frac{1}{25}\left(\begin{smallmatrix}9 & 12\\ 12 & 16\end{smallmatrix}\right)$, so $2uu^T = \frac{1}{25}\left(\begin{smallmatrix}18 & 24\\ 24 & 32\end{smallmatrix}\right)$. Subtracting $I = \frac{1}{25}\left(\begin{smallmatrix}25 & 0\\ 0 & 25\end{smallmatrix}\right)$ gives $\frac{1}{25}\left(\begin{smallmatrix}-7 & 24\\ 24 & 7\end{smallmatrix}\right)$. *Check:* it fixes $u$, since $\frac{1}{25}\cdot\frac{1}{5}(-21+96,\ 72+28) = \frac{1}{125}(75, 100) = (\frac{3}{5}, \frac{4}{5})$. Its trace is 0 and its determinant is $-1$, as for every reflection in the plane.

**Why the other options are incorrect.**

**(A)** This is $2uu^T$ alone: the $I$ was not subtracted, and the matrix is not orthogonal.

**(B)** The off-diagonal entries are those of $uu^T$ and were not doubled; the columns are then not unit vectors.

**(D)** This is $I - 2uu^T$, the negative of the required matrix; it sends $u$ to $-u$.

**Key takeaway.** Check a reflection by testing that it fixes $u$, and that its trace is 0 and determinant $-1$ in the plane.

**Q.3** SINGLE-CORRECT MCQ · Two reflections make a rotation

A vector in the plane is reflected first in the line through the origin at angle $10^\circ$ to the $x$-axis, and then in the line through the origin at angle $35^\circ$. The combined map is

**(A)** a reflection in the line at angle $45^\circ$ **(B)** a rotation by $+25^\circ$

**(C)** a rotation by $-50^\circ$ **(D)** ✓ a rotation by $+50^\circ$

**Correct answer: (D)** **Detailed solution**

**Concepts required.** Reflection in the line at angle $\alpha$: $\phi \mapsto 2\alpha - \phi$. Two reflections compose to a rotation by twice the angle between the lines.

**Relevant formulae.** A reflection in the line at angle $\alpha$ maps angle $\phi$ to $2\alpha - \phi$. Reflecting at $\alpha$ and then at $\beta$ rotates by $2(\beta - \alpha)$.

**Step-by-step explanation.** A reflection in the line at angle $\alpha$ sends a vector at angle $\phi$ to angle $2\alpha - \phi$. Applying $\alpha = 10^\circ$ and then $\beta = 35^\circ$:

$$\phi \;\to\; 20^\circ - \phi \;\to\; 70^\circ - (20^\circ - \phi) = \phi + 50^\circ.$$

The result is a rotation by $2(\beta - \alpha) = +50^\circ$: **twice** the angle between the lines, in the direction from the first line to the second. This is the geometric heart of Grover's algorithm.

**Why the other options are incorrect.**

**(A)** Two reflections have determinant $(-1)(-1) = +1$, so the product is a rotation and cannot be a reflection.

**(B)** $25^\circ$ is the angle between the lines; the rotation angle is twice this.

**(C)** $-50^\circ$ is the result for the opposite order (first $35^\circ$, then $10^\circ$).

**Key takeaway.** Two reflections make a rotation by twice the angle between their axes: this is why $G$ rotates by $2\theta$.

**Q.4** SINGLE-CORRECT MCQ · Eigenvalues of a rotation

Let $\sin\theta = \frac{1}{4}$ with $0 < \theta < \frac{\pi}{2}$, and let $G = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}$. The eigenvalues of $G$ are

**(A)** $\dfrac{7 \pm \sqrt{15}}{8}$ **(B)** $\dfrac{1 \pm i\sqrt{15}}{4}$ **(C)** $\pm 1$ **(D)** ✓ $\dfrac{7 \pm i\sqrt{15}}{8}$

**Correct answer: (D)** **Detailed solution**

**Concepts required.** A plane rotation by $\alpha$ has eigenvalues $e^{\pm i\alpha}$, and $\cos 2\theta = 1 - 2\sin^2\theta$.

**Relevant formulae.** A rotation by $\alpha$ has eigenvalues $\lambda_\pm = e^{\pm i\alpha}$; $\cos 2\theta = 1 - 2\sin^2\theta$; $\sin 2\theta = 2\sin\theta\cos\theta$.

**Step-by-step explanation.** A rotation by angle $\alpha$ has eigenvalues $e^{\pm i\alpha} = \cos\alpha \pm i\sin\alpha$ (trace $2\cos\alpha$, determinant 1). Here $\alpha = 2\theta$, with

$$\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{2}{16} = \frac{7}{8}, \qquad \sin 2\theta = 2 \cdot \frac{1}{4} \cdot \frac{\sqrt{15}}{4} = \frac{\sqrt{15}}{8}.$$

Therefore the eigenvalues are $\frac{7 \pm i\sqrt{15}}{8}$; their modulus is $\sqrt{(49 + 15)/64} = 1$, as required.

**Why the other options are incorrect.**

**(A)** These real numbers are not of modulus 1; the factor $i$ was lost.

**(B)** These are $e^{\pm i\theta}$, using $\theta$ instead of the rotation angle $2\theta$.

**(C)** $\pm 1$ are the eigenvalues of a reflection, or of a rotation by 0 or $\pi$ only.

**Key takeaway.** Eigenvalues of a rotation lie on the unit circle at the rotation angle: quantum counting reads $\theta$ from them.

**Q.5** SINGLE-CORRECT MCQ · Powers of a complex phase

Let $z = \dfrac{1+i}{\sqrt{2}}$. The value of $z^6$ is

**(A)** ✓ $-i$ **(B)** $-1$ **(C)** $1$ **(D)** $i$

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Polar form $z = |z|e^{i\varphi}$, and $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$.
**Relevant formulae.** $z = |z|e^{i\varphi}$ with modulus $|z|$ and angle $\varphi$; $z^m = |z|^m e^{im\varphi}$; $e^{i\varphi} = \cos\varphi + i\sin\varphi$.
**Step-by-step explanation.** Write $z$ in polar form: $|z| = \sqrt{\frac{1}{2}+\frac{1}{2}} = 1$ and its angle is $45°$, so $z = e^{i\pi/4}$. Powers multiply the angle:

$$z^6 = e^{6i\pi/4} = e^{3i\pi/2} = \cos 270° + i\sin 270° = -i.$$

Since $|z| = 1$, multiplying an amplitude by $z$ changes only its phase and never its probability.
**Why the other options are incorrect.**
**(B)** $-1 = e^{i\pi}$ is $z^4$.
**(C)** 1 would need a whole number of turns; $z^8 = 1$, but $z^6$ has turned only $270°$.
**(D)** $i = e^{i\pi/2}$ corresponds to $z^2$ or a sign slip in the angle $270°$.
**Key takeaway.** Convert to polar form first: powers of a phase simply add angles.

**Q.6** SINGLE-CORRECT MCQ · Optimal integer $k$; why $-\frac{1}{2}$ matters

Let $\sin\theta = \frac{1}{10}$, $0 < \theta < \frac{\pi}{2}$; numerically $\theta = 0.10017$. Among the integers $0 \le k \le 10$, the one that makes $\sin^2\left((2k+1)\theta\right)$ largest is

**(A)** 8 **(B)** 10 **(C)** 5 **(D)** ✓ 7

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $\sin^2((2k+1)\theta)$ is largest when $(2k+1)\theta$ is closest to $\frac{\pi}{2}$, that is for $k$ nearest to $\frac{\pi}{4\theta} - \frac{1}{2}$.
**Relevant formulae.** $\sin^2\alpha$ is largest at $\alpha = \frac{\pi}{2}$; so $(2k+1)\theta = \frac{\pi}{2}$ gives the real value $k = \frac{\pi}{4\theta} - \frac{1}{2}$, to be rounded to an integer.
**Step-by-step explanation.** We want $(2k+1)\theta$ as close as possible to $\frac{\pi}{2}$, i.e. $k$ closest to $\frac{\pi}{4\theta} - \frac{1}{2}$. Here

$$\frac{\pi}{4\theta} - \frac{1}{2} = \frac{3.1416}{0.40067} - 0.5 = 7.841 - 0.5 = 7.341,$$

so $k = 7$: $15\theta = 1.5025$ rad is 0.068 below $\frac{\pi}{2}$, while $17\theta = 1.7028$ rad is 0.132 above it. Then $\sin^2 15\theta = 0.9953$ and $\sin^2 17\theta = 0.9827$. The tempting estimate $\frac{\pi}{4}\cdot 10 = 7.85$ rounds to 8, which is wrong: it omits the $-\frac{1}{2}$.
**Why the other options are incorrect.**
**(A)** $k = 8$ comes from rounding $\frac{\pi}{4}\sqrt{N} = 7.85$; it overshoots, and $\sin^2 17\theta = 0.983 < 0.995$.
**(B)** $k = 10 = \sqrt{N}$ omits the factor $\frac{\pi}{4}$, giving $21\theta = 2.10$ rad and $\sin^2 \approx 0.74$.
**(C)** $k = 5$ stops too soon: $11\theta = 1.10$ rad and $\sin^2 \approx 0.79$.
**Key takeaway.** Use $\frac{\pi}{4\theta} - \frac{1}{2}$, not $\frac{\pi}{4}\sqrt{N}$, when the exact integer matters.

**Q.7** SINGLE-CORRECT MCQ · Inversion about the mean

The vector $v \in \mathbb{R}^8$ has seven entries equal to $+1$ and one entry equal to $-1$. Each entry $v_j$ is replaced by $2\bar{v} - v_j$, where $\bar{v}$ is the mean of the entries. The new value of the entry that was $-1$ is

**(A)** ✓ $\frac{5}{2}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{7}{4}$ **(D)** $\frac{1}{2}$

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Inversion about the mean: $v_j \mapsto 2\bar{v} - v_j$. It preserves the length of $v$ and leaves the mean unchanged.

**Relevant formulae.** Mean $\bar{v} = \frac{1}{N}\sum_j v_j$ of the $N$ entries; inversion about the mean: $v_j \mapsto 2\bar{v} - v_j$.

**Step-by-step explanation.** The mean is $\bar{v} = \frac{7-1}{8} = \frac{3}{4}$. The special entry becomes $2\cdot\frac{3}{4} - (-1) = \frac{5}{2}$; each other entry becomes $\frac{3}{2} - 1 = \frac{1}{2}$. *Check:* the length is preserved, $7\cdot\frac{1}{4} + \frac{25}{4} = 8 = 7 + 1$. Inversion about the mean reflects every entry through the mean, so an entry far below the mean ends up far above it.

**Why the other options are incorrect.**

**(B)** $\frac{3}{2} = 2\bar{v}$ forgets the term $-v_j = +1$.

**(C)** $\frac{7}{4}$ comes from using the mean $\frac{7}{8}$ of a wrong sum and is not consistent with length preservation.

**(D)** $\frac{1}{2}$ is the new value of the *other* entries.

**Key takeaway.** An entry a distance $d$ below the mean ends a distance $d$ above it. Always check that the length is preserved.

**Q.8** SINGLE-CORRECT MCQ · Classical search without replacement

A box holds 31 cards of which exactly 3 are marked. Cards are drawn one at a time at random without replacement until the first marked card appears. The expected number of draws is

**(A)** $\frac{31}{4}$ **(B)** $\frac{31}{3}$ **(C)** ✓ 8 **(D)** 16

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Without replacement, the expected position of the first of $M$ marked cards among $N$ is $\frac{N+1}{M+1}$ (symmetry of the gaps).

**Relevant formulae.** Expected draws without replacement: $\frac{N+1}{M+1}$, with $N$ items of which $M$ are marked. With replacement: $\frac{N}{M}$.

**Step-by-step explanation.** The $M = 3$ marked cards cut the random order of the $N - M = 28$ unmarked cards into $M + 1 = 4$ gaps, and by symmetry each gap holds on average $\frac{28}{4} = 7$ unmarked cards. The first marked card therefore comes after 7 unmarked ones on average, on draw number 8. In general the expected number of draws is $\frac{N+1}{M+1} = \frac{32}{4} = 8$. The value $\frac{31}{3}$ would be right for drawing *with* replacement.

**Why the other options are incorrect.**

**(A)** $\frac{31}{4} = \frac{N}{M+1}$ uses $N$ where $N + 1$ is needed.

**(B)** $\frac{31}{3} = N/M$ is the expectation *with* replacement (geometric distribution).

**(D)** $16 = \frac{N+1}{2}$ is the formula for one marked card.

**Key takeaway.** With $M = 1$ the formula gives the classical average $\frac{N+1}{2}$ that Grover is compared with.

**Q.9** Single-correct MCQ · Binary vs. linear search

A key, which may be absent, is looked up in a list of 1023 distinct entries using classical comparisons. The worst-case numbers of comparisons are, for binary search on the *sorted* list and for linear search on the *unsorted* list, respectively,

**(A)** ✓ 10 and 1023 **(B)** 1023 and 1023 **(C)** 10 and 512 **(D)** 32 and 1023

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Binary search halves a sorted range at each step, so it needs about $\log_2 N$ comparisons. Without order, nothing can be excluded without looking at it.

**Relevant formulae.** Binary search on $N = 2^b - 1$ sorted entries: $b$ comparisons at most. Linear search on $N$ unsorted entries: $N$ comparisons at most.

**Step-by-step explanation.** Each comparison in binary search halves the remaining range. $1023 = 2^{10} - 1$ entries are exhausted after exactly 10 comparisons. An unsorted list has no structure to exploit: if the key is absent or in the last place, all 1023 entries must be checked. Grover's algorithm addresses the second, *unstructured*, case. (512 is about the *average* cost of a linear search when the key is present.)

**Why the other options are incorrect.**

**(B)** 1023 for the sorted list ignores the order, which is exactly what binary search exploits.

**(C)** 512 is about the average cost of a linear search when the key is present, not the worst case.

**(D)** $32 \approx \sqrt{1023}$ is a square-root cost, which no classical comparison search on a sorted list needs.

**Key takeaway.** Structure gives $\log_2 N$. Without structure the classical cost is $N$; this is the setting of Grover's algorithm.

**Q.10** Single-correct MCQ · Counting solutions of a Boolean function

Let $f(x_1, x_2, x_3, x_4) = (x_1 \vee x_2) \wedge (x_3 \oplus x_4)$ on $\{0,1\}^4$. The fraction of inputs with $f = 1$ is

**(A)** $\frac{1}{2}$ **(B)** ✓ $\frac{3}{8}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{5}{8}$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Count the solutions of a Boolean function. If factors share no variables, the counts multiply. $\sin^2\theta = M/N$.

**Relevant formulae.** Fraction of solutions $= M/N$, with $M$ inputs satisfying $f$ out of $N = 2^n$. Conditions on disjoint variables: counts multiply.

**Step-by-step explanation.** The two factors involve different variables, so the counts multiply. $x_1 \vee x_2 = 1$ for 3 of the 4 pairs $(x_1, x_2)$; $x_3 \oplus x_4 = 1$ for 2 of the 4 pairs $(x_3, x_4)$. So $M = 3 \times 2 = 6$ of $N = 16$ inputs, a fraction $\frac{6}{16} = \frac{3}{8}$. This fraction is the quantity $\sin^2\theta$ in a Grover search for the solutions of $f$.

**Why the other options are incorrect.**

**(A)** $\frac{1}{2}$ is the fraction for $x_3 \oplus x_4$ alone.

**(C)** $\frac{1}{4} = \frac{4}{16}$ counts only two solutions of $x_1 \vee x_2$ instead of three.

**(D)** $\frac{5}{8}$ averages the fractions $\frac{3}{4}$ and $\frac{1}{2}$ instead of multiplying them.

**Key takeaway.** Find $M$ by factoring the Boolean expression into independent parts.

**Q.11** Single-correct MCQ · The signed sum $N - 2M$

A Boolean function $f$ on 4 bits takes the value 1 on exactly 3 inputs. The value of $\sum_{x\in\{0,1\}^4} (-1)^{f(x)}$ is

**(A)** ✓ 10 **(B)** 7 **(C)** 6 **(D)** 13

**Correct answer: (A)** **Detailed solution**

**Concepts required.** $(-1)^{f(x)}$ is $+1$ on the $N - M$ unmarked inputs and $-1$ on the $M$ marked ones, so the sum is $N - 2M$.
**Relevant formulae.** $\sum_x (-1)^{f(x)} = (N - M) - M = N - 2M$, with $M$ inputs having $f(x) = 1$.
**Step-by-step explanation.** Each of the $N - M = 13$ inputs with $f = 0$ contributes $+1$, and each of the $M = 3$ inputs with $f = 1$ contributes $-1$:

$$\sum_x (-1)^{f(x)} = (N - M) - M = N - 2M = 16 - 6 = 10.$$

Divided by $N$ this is the factor $1 - \frac{2M}{N}$ that appears in the mean amplitude after the oracle.
**Why the other options are incorrect.**
**(B)** 7 is $N/2 - 1$ and does not follow from the counts.
**(C)** $6 = 2M$ is the amount lost from $N$, not the sum.
**(D)** $13 = N - M$ counts the unmarked inputs and ignores the $-1$ from the marked ones.
**Key takeaway.** The mean amplitude after the oracle is proportional to $N - 2M$.

**Q.12** Single-correct MCQ · Normalising a state vector

The vector $|\psi\rangle = c\,(\,|00\rangle + i\,|01\rangle + (1+i)\,|10\rangle + 2\,|11\rangle\,)$ with $c > 0$ is normalised. The value of $c$ is

**(A)** $\frac{1}{5}$ **(B)** ✓ $\frac{1}{2\sqrt{2}}$ **(C)** $\frac{1}{8}$ **(D)** $\frac{1}{\sqrt{7}}$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Normalisation: $\sum_x |c_x|^2 = 1$, with $|a + ib|^2 = a^2 + b^2$.
**Relevant formulae.** Normalisation: $\sum_x |c_x|^2 = 1$ for amplitudes $c_x$; $|a + ib|^2 = a^2 + b^2$ for real $a, b$.
**Step-by-step explanation.** Normalisation means $\sum_x |c_x|^2 = 1$, where each term is a squared *modulus*: $|1|^2 = 1$, $|i|^2 = 1$, $|1 + i|^2 = 1^2 + 1^2 = 2$, $|2|^2 = 4$. So

$$c^2(1 + 1 + 2 + 4) = 8c^2 = 1 \;\Rightarrow\; c = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}}.$$

**Why the other options are incorrect.**
**(A)** $\frac{1}{5}$ adds non-squared sizes such as $1 + 1 + 1 + 2$ instead of squared moduli.
**(C)** $\frac{1}{8}$ forgets the square root in $c^2 = \frac{1}{8}$.
**(D)** $\frac{1}{\sqrt{7}}$ takes $|1 + i|^2 = 1$ instead of 2.
**Key takeaway.** Use squared moduli, never squares of the complex numbers themselves.

**Q.13** Single-correct MCQ · Square-root scaling

Let $g(n) = \frac{\pi}{4}\sqrt{2^n}$. Then $g(n+6)/g(n)$ equals

**(A)** 6 **(B)** ✓ 8 **(C)** $\sqrt{6}$ **(D)** 64

**Correct answer: (B)** **Detailed solution**

**Concepts required.** $\sqrt{2^n} = 2^{n/2}$, so adding $a$ bits multiplies $\sqrt{N}$ by $2^{a/2}$.
**Relevant formulae.** $\sqrt{2^n} = 2^{n/2}$, so $\frac{\sqrt{2^{n+a}}}{\sqrt{2^n}} = 2^{a/2}$ when $a$ bits are added.
**Step-by-step explanation.** $\frac{g(n+6)}{g(n)} = \sqrt{\frac{2^{n+6}}{2^n}} = \sqrt{2^6} = 2^3 = 8$. Six extra bits make the search space 64 times larger but the number of Grover iterations only 8 times larger.
**Why the other options are incorrect.**
**(A)** 6 treats the cost as linear in the number of bits.
**(C)** $\sqrt{6}$ takes the root of the *number of added bits* rather than of $2^6$.
**(D)** $64 = 2^6$ is the growth of $N$, which is the classical scaling.
**Key takeaway.** Each extra qubit multiplies the iteration count by $\sqrt{2}$.

**Q.14** Single-correct MCQ · Complex inner product

Let $u = \frac{1}{\sqrt{2}}(1, i, 0)^T$ and $v = \frac{1}{\sqrt{3}}(1, 1, 1)^T$. The inner product $\langle u|v\rangle$ is

**(A)** $\frac{1+i}{\sqrt{6}}$ **(B)** $\frac{2}{\sqrt{6}}$ **(C)** ✓ $\frac{1-i}{\sqrt{6}}$ **(D)** 0

**Correct answer: (C)** **Detailed solution**

**Concepts required.** $\langle u|v\rangle = \sum_k \overline{u_k} v_k$: conjugate the first vector.
**Relevant formulae.** $\langle u|v\rangle = \sum_k \overline{u_k} v_k$, where the bar means the complex conjugate.
**Step-by-step explanation.** $\langle u|v\rangle = \sum_k \overline{u_k} v_k$: the *first* vector is conjugated. Thus

$$\langle u|v\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}}(\bar{1} \cdot 1 + \bar{i} \cdot 1 + 0) = \frac{1-i}{\sqrt{6}}.$$

Forgetting the conjugate gives $\frac{1+i}{\sqrt{6}}$, which is $\langle v|u\rangle$. The probability $|\langle u|v\rangle|^2 = \frac{2}{6} = \frac{1}{3}$ is the same either way.
**Why the other options are incorrect.**
**(A)** $\frac{1+i}{\sqrt{6}}$ forgets the conjugate on $u$; it is $\langle v|u\rangle$.
**(B)** $\frac{2}{\sqrt{6}}$ uses the moduli of the components and loses the phase $i$.
**(D)** 0 would mean orthogonal vectors; the first component alone gives a non-cancelling term.
**Key takeaway.** Swapping the two vectors conjugates the inner product: $\langle v|u\rangle = \overline{\langle u|v\rangle}$.

**Q.15** MATCHING LISTS · $\sin^2 3\theta$ as a polynomial

Match the value of $\sin^2\theta$ (List-I) with the value of $\sin^2 3\theta$ (List-II).

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | $\sin^2\theta = \frac{1}{8}$ | **(1)** | 1 |
| **(Q)** | $\sin^2\theta = \frac{1}{4}$ | **(2)** | 0 |
| **(R)** | $\sin^2\theta = \frac{1}{2}$ | **(3)** | $\frac{25}{32}$ |
| **(S)** | $\sin^2\theta = \frac{3}{4}$ | **(4)** | $\frac{1}{2}$ |

**(A)** P→4, Q→1, R→3, S→2
**(B)** P→3, Q→1, R→2, S→4
**(C)** ✓ P→3, Q→1, R→4, S→2
**(D)** P→1, Q→3, R→4, S→2

**Correct answer: (C)** **Detailed solution**

**Concepts required.** $\sin^2 3\theta = s(3-4s)^2$ with $s = \sin^2\theta$.
**Relevant formulae.** $\sin^2 3\theta = s(3-4s)^2$ with $s = \sin^2\theta$.
**Step-by-step explanation.** Since $\sin 3\theta = \sin\theta\,(3 - 4\sin^2\theta)$, with $s = \sin^2\theta$ we get $\sin^2 3\theta = s(3-4s)^2$. $s = \frac{1}{8}$: $\frac{1}{8}\cdot\frac{25}{4} = \frac{25}{32}$ (3). $s = \frac{1}{4}$: $\frac{1}{4}\cdot 4 = 1$ (1). $s = \frac{1}{2}$: $\frac{1}{2}\cdot 1 = \frac{1}{2}$ (4). $s = \frac{3}{4}$: $\frac{3}{4}\cdot 0 = 0$ (2). Hence P→3, Q→1, R→4, S→2.
**Why the other options are incorrect.**
**(A)** This gives $s = \frac{1}{8}$ the value $\frac{1}{2}$, but $\frac{1}{8}\cdot\frac{25}{4} = \frac{25}{32}$.
**(B)** This exchanges R and S; for $s = \frac{3}{4}$ we have $3 - 4s = 0$, so the value is 0.
**(D)** This exchanges P and Q; for $s = \frac{1}{4}$ the value is 1, not $\frac{25}{32}$.
**Key takeaway.** $s = \frac{1}{4}$ is perfect, $s = \frac{1}{2}$ is flat and $s = \frac{3}{4}$ is the worst case for one iteration.

**Q.16** MATCHING LISTS · Traces of projectors and reflections

Let $|u\rangle$ be a unit vector in $\mathbb{C}^N$, with $N \geq 3$. Match each matrix in List-I with its trace in List-II.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | $2\,\lvert u\rangle\langle u\rvert - I$ | **(1)** | $N$ |
| **(Q)** | $I - 2\,\lvert u\rangle\langle u\rvert$ | **(2)** | $N-2$ |
| **(R)** | $\lvert u\rangle\langle u\rvert$ | **(3)** | 1 |
| **(S)** | $I$ | **(4)** | $2-N$ |

**(A)** P→3, Q→2, R→4, S→1
**(B)** P→4, Q→2, R→1, S→3
**(C)** P→2, Q→4, R→3, S→1
**(D)** ✓ P→4, Q→2, R→3, S→1

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $\operatorname{tr}|u\rangle\langle u| = \langle u|u\rangle = 1$ and trace is linear.
**Relevant formulae.** $\operatorname{tr}|u\rangle\langle u| = \langle u|u\rangle = 1$; $\operatorname{tr} I = N$ in dimension $N$; trace is linear.
**Step-by-step explanation.** The key fact is $\operatorname{tr}|u\rangle\langle u| = \langle u|u\rangle = 1$, and $\operatorname{tr} I = N$. So $\operatorname{tr}(2\,|u\rangle\langle u| - I) = 2 - N$ (4); $\operatorname{tr}(I - 2\,|u\rangle\langle u|) = N - 2$ (2); $\operatorname{tr}|u\rangle\langle u| = 1$ (3); $\operatorname{tr} I = N$ (1). Hence P→4, Q→2, R→3, S→1.
**Why the other options are incorrect.**
**(A)** This gives $2\,|u\rangle\langle u| - I$ trace 1 and the projector trace $2 - N$, which ignores linearity.
**(B)** This swaps R and S: the projector has trace 1, the identity has trace $N$.

**(C)** This swaps P and Q: the sign in front of the identity decides which of $N-2$, $2-N$ applies.
**Key takeaway.** Trace identifies eigenvalue multiplicities: $D$ has one $+1$ and $N-1$ eigenvalues $-1$.

**Q.17** Assertion–Reason · Rounding rule vs. $\arcsin x > x$

**Assertion (A):** Rounding $\frac{\pi}{4}\sqrt{N}$ to the nearest integer does not always give the integer $k$ for which $(2k+1)\theta$ is closest to $\frac{\pi}{2}$, where $\sin\theta = 1/\sqrt{N}$.

**Reason (R):** For $0 < x < 1$ we have $\arcsin x < x$.

**(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
**(B)** Both A and R are true, but R is *not* the correct explanation of A
**(C)** ✓ A is true, but R is false
**(D)** A is false, but R is true

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Decide A and R independently. $\arcsin x = x + x^3/6 + \ldots$, so it exceeds $x$ for $x > 0$.
**Relevant formulae.** $k^* = \frac{\pi}{4\theta} - \frac{1}{2}$; $\arcsin x = x + \frac{x^3}{6} + \ldots$ for $|x| < 1$.
**Step-by-step explanation.** **A is true.** For $N = 100$, $\frac{\pi}{4}\sqrt{N} = 7.85$ rounds to 8, but the true optimum is $k = 7$ (see Q.6). **R is false:** the inequality goes the other way. $\arcsin x = x + \frac{x^3}{6} + \cdots > x$, for example $\arcsin 0.1 = 0.10017$. The actual reason for A is the omitted $-\frac{1}{2}$ in $k = \frac{\pi}{4\theta} - \frac{1}{2}$. Answer: (C).
**Why the other options are incorrect.**
**(A)** R is false, so it cannot explain A.
**(B)** This needs R to be true, but the inequality in R points the wrong way.
**(D)** A is true: $N = 100$ is a counterexample to the rounding rule, and R is false.
**Key takeaway.** An Assertion can be true while the Reason offered for it is false: test both with a number.

**Q.18** Statement I / II · Spectrum of a reflection

Let $|u\rangle$ be a unit vector in $\mathbb{C}^N$, with $N \geq 2$, and let $D = 2|u\rangle\langle u| - I$.

**Statement I:** $D^2 = I$.

**Statement II:** $D$ has the eigenvalue $+1$ with multiplicity 1 and the eigenvalue $-1$ with multiplicity $N-1$.

**(A)** ✓ Both Statement I and Statement II are true
**(B)** Both Statement I and Statement II are false
**(C)** Statement I is true, but Statement II is false
**(D)** Statement I is false, but Statement II is true

**Correct answer: (A)** **Detailed solution**

**Concepts required.** $D$ is a reflection: $D^2 = I$, it fixes $|u\rangle$, and it negates the orthogonal complement.
**Relevant formulae.** $D = 2|u\rangle\langle u| - I$ with $\langle u|u\rangle = 1$; $D|u\rangle = |u\rangle$; $D|w\rangle = -|w\rangle$ if $\langle u|w\rangle = 0$.
**Step-by-step explanation.** **I is true:** $D^2 = 4|u\rangle\langle u|u\rangle\langle u| - 4|u\rangle\langle u| + I = I$, because $\langle u|u\rangle = 1$. **II is true:** $D|u\rangle = 2|u\rangle - |u\rangle = |u\rangle$, and for any $|w\rangle$ orthogonal to $|u\rangle$ we have $D|w\rangle = -|w\rangle$; such vectors span an $(N-1)$-dimensional space. *Check:* the trace is $1 - (N-1) = 2 - N$, in agreement with Q.16. Answer: (A).
**Why the other options are incorrect.**

**(B)** Both are true: $\langle u|u\rangle = 1$ gives I, and acting on $|u\rangle$ and on its complement gives II.

**(C)** II is also true, as the trace check $1-(N-1) = 2-N$ confirms.

**(D)** I is true by direct expansion.

**Key takeaway.** Every reflection is its own inverse and has only eigenvalues $\pm 1$.

## Paragraph for Questions 19 and 20

Define $p(k) = \sin^2\left((2k+1)\theta\right)$ for integers $k \geq 0$, with $\theta = \pi/18 = 10°$. (This is the shape of the Grover success curve, but the two questions below need only trigonometry.) Note that $(2k+1)\theta$ takes the values $10°, 30°, 50°, \ldots$

**Q.19** PARAGRAPH BASED · First maximum and period of $p(k)$

The smallest $k$ with $p(k) = 1$, and the smallest positive period of the sequence $p(k)$, are

**(A)** first maximum at $k = 4$; smallest period 18

**(B)** first maximum at $k = 5$; smallest period 9

**(C)** first maximum at $k = 9$; smallest period 18

**(D)** ✓ first maximum at $k = 4$; smallest period 9

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $p(k) = 1$ requires an odd multiple of 90°. The period comes from $2\theta$ steps compared with 180°.

**Relevant formulae.** $p(k) = \sin^2((2k+1)\theta)$; $\sin^2$ has period 180°; moving $k$ by $q$ adds $2q\theta$ to the angle.

**Step-by-step explanation.** $p(k) = 1$ requires $(2k+1)\cdot 10° = 90°$ (the first odd multiple of 90°), i.e. $2k+1 = 9$, so $k = 4$. Moving from $k$ to $k+q$ adds $20°q$ to the angle. $\sin^2$ has period 180°, so we need $20°q = 180°m$; the smallest solution is $q = 9$. Hence period 9 ($q = 18$ corresponds to the period 360° of sin and is not the smallest).

**Why the other options are incorrect.**

**(A)** 18 is a period but not the smallest: it corresponds to 360°.

**(B)** $k = 5$ gives 110°, already past the maximum.

**(C)** $k = 9$ gives 190°, which is near a minimum, and 18 is not the smallest period.

**Key takeaway.** Convert to degrees and look at the list of angles: the sequence becomes easy to read.

**Q.20** PARAGRAPH BASED · Tolerance window around the peak

The number of integers $k$ with $0 \leq k \leq 17$ for which $p(k) \geq \frac{3}{4}$ is

**(A)** 3 **(B)** ✓ 6 **(C)** 9 **(D)** 4

**Correct answer: (B)** **Detailed solution**

**Concepts required.** $\sin^2\alpha \geq \frac{3}{4} \iff |\sin\alpha| \geq \frac{\sqrt{3}}{2}$; $\alpha$ is within 30° of an odd multiple of 90°.

**Relevant formulae.** $\sin^2\alpha \geq \frac{3}{4} \iff |\sin\alpha| \geq \frac{\sqrt{3}}{2}$, i.e. $\alpha$ is within 30° of an odd multiple of 90°.

**Step-by-step explanation.** $\sin^2\alpha \geq \frac{3}{4}$ means $|\sin\alpha| \geq \frac{\sqrt{3}}{2}$, i.e. $\alpha$ lies within 30° of 90° or of 270°. For $k = 0, \ldots, 17$ the angle $(2k+1)\cdot 10°$ runs over $10°, 30°, \ldots, 350°$. The values in $[60°, 120°]$ are $70°, 90°, 110°$ ($k = 3, 4, 5$); the values in $[240°, 300°]$ are $250°, 270°, 290°$ ($k = 12, 13, 14$). That makes 6 values of $k$.

**Why the other options are incorrect.**

**(A)** 3 counts only one of the two peaks in the range $0 \leq k \leq 17$.

**(C)** 9 is the period and not a count of angles in the window.

**(D)** 4 counts two values at each peak; each peak has three: $70°, 90°, 110°$ and $250°, 270°, 290°$.

**Key takeaway.** A window of $\pm 30°$ around the peak corresponds to $P \geq \frac{3}{4}$: there is some tolerance around $k_{\text{opt}}$.

## SECTION B · Quantum Computing Foundations Q.21–40

**Q.21** SINGLE-CORRECT MCQ · Uniform state, partial outcome

The state $H^{\otimes 3}|000\rangle$ is measured in the computational basis. The probability that qubit 1 gives 1 and qubit 3 gives 0 is

**(A)** $\frac{1}{8}$ **(B)** ✓ $\frac{1}{4}$ **(C)** $\frac{1}{\sqrt{8}}$ **(D)** $\frac{1}{2}$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** $H^{\otimes n}|0\ldots0\rangle$ is uniform and a product state, so its qubits are independent.
**Relevant formulae.** $H^{\otimes n}|0\ldots0\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle$ with $N = 2^n$; Born rule: $P(x) = |c_x|^2$; the probability of an event is the sum over its outcomes.
**Step-by-step explanation.** All 8 outcomes have probability $\left|\frac{1}{\sqrt{8}}\right|^2 = \frac{1}{8}$. The event "$x_1 = 1$, $x_3 = 0$" contains the two outcomes $|100\rangle$ and $|110\rangle$, so $P = \frac{2}{8} = \frac{1}{4}$. (Equivalently the qubits are independent: $\frac{1}{2}\cdot\frac{1}{2}$.)
**Why the other options are incorrect.**
**(A)** $\frac{1}{8}$ is the probability of one specific full string; here qubit 2 is free.
**(C)** $\frac{1}{\sqrt{8}}$ is an amplitude, not a probability.
**(D)** $\frac{1}{2}$ is the probability for one qubit only.
**Key takeaway.** Sum over all full strings consistent with a partial outcome.

**Q.22** SINGLE-CORRECT MCQ · Tensor-product amplitude

Let $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The amplitude of $|10\rangle$ in $|\psi\rangle$ is

**(A)** ✓ $\frac{i}{2}$ **(B)** $-\frac{i}{2}$ **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{2}$

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Tensor product: the amplitude of $|jk\rangle$ is $a_j b_k$, with qubit 1 written first.
**Relevant formulae.** $(a_0|0\rangle + a_1|1\rangle) \otimes (b_0|0\rangle + b_1|1\rangle) = \sum_{j,k} a_j b_k |jk\rangle$.
**Step-by-step explanation.** Multiply out: $(a_0|0\rangle + a_1|1\rangle) \otimes (b_0|0\rangle + b_1|1\rangle)$ has the amplitude $a_j b_k$ on $|jk\rangle$. For $|10\rangle$: $a_1 b_0 = \frac{i}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}} = \frac{i}{2}$. The whole state is $\frac{1}{2}(|00\rangle - |01\rangle + i|10\rangle - i|11\rangle)$.
**Why the other options are incorrect.**
**(B)** $-\frac{i}{2}$ is the amplitude of $|11\rangle$.
**(C)** $\frac{1}{2}$ is the amplitude of $|00\rangle$.
**(D)** $-\frac{1}{2}$ is the amplitude of $|01\rangle$.
**Key takeaway.** Read the first bit from the first factor and the second bit from the second factor.

**Q.23** Single-correct MCQ · Hadamard layer signs

$H \otimes H$ is applied to $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$. The amplitude of $|11\rangle$ in the result is

**(A)** $\frac{1}{2}$ **(B)** $0$ **(C)** ✓ $-\frac{1}{2}$ **(D)** $1$

**Correct answer: (C) Detailed solution**

**Concepts required.** $H^{\otimes 2}$ has entries $\frac{1}{2}(-1)^{x\cdot y}$; each output amplitude is a signed sum of the input amplitudes.
**Relevant formulae.** $H^{\otimes n}|x\rangle = \frac{1}{\sqrt{N}}\sum_y (-1)^{x\cdot y}|y\rangle$, with $x \cdot y = \sum_j x_j y_j$.
**Step-by-step explanation.** Use $H^{\otimes 2}|x\rangle = \frac{1}{2}\sum_y (-1)^{x\cdot y}|y\rangle$. For $y = 11$ the signs $(-1)^{x\cdot y}$ for $x = 00, 01, 10, 11$ are $+, -, -, +$, so the amplitude is

$$\frac{1}{2}\cdot\frac{1}{2}[(+1)(1) + (-1)(1) + (-1)(1) + (+1)(-1)] = \frac{1}{4}(1 - 1 - 1 - 1) = -\frac{1}{2}.$$

The full result is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$: this state is unchanged by $H \otimes H$.
**Why the other options are incorrect.**
**(A)** $\frac{1}{2}$ has the wrong sign, from treating the entry for $x = 11$, $y = 11$ as negative.
**(B)** $0$ would need the signed sum to cancel; here it is $-2$ before the factor $\frac{1}{4}$.
**(D)** $1$ is the magnitude when all four terms align, which they do not.
**Key takeaway.** Compute one output amplitude at a time with $(-1)^{x\cdot y}$.

**Q.24** Single-correct MCQ · Phase kickback with $|-\rangle$

Let $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Then $U_f|x\rangle|-\rangle$ equals

**(A)** $|x\rangle|f(x)\rangle$ **(B)** $|x\rangle|-\rangle$ **(C)** ✓ $(-1)^{f(x)}|x\rangle|-\rangle$ **(D)** $|f(x)\rangle|-\rangle$

**Correct answer: (C) Detailed solution**

**Concepts required.** Phase kickback: an ancilla in $|-\rangle$ is an eigenvector of $X$ with eigenvalue $-1$.
**Relevant formulae.** $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$, with $\oplus$ the XOR; $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$; $X|-\rangle = -|-\rangle$.
**Step-by-step explanation.** Expand the ancilla: $U_f|x\rangle|-\rangle = \frac{1}{\sqrt{2}}|x\rangle(|f(x)\rangle - |1 \oplus f(x)\rangle)$. If $f(x) = 0$ the bracket is $|0\rangle - |1\rangle$; if $f(x) = 1$ it is $|1\rangle - |0\rangle = -(|0\rangle - |1\rangle)$. So the result is $(-1)^{f(x)}|x\rangle|-\rangle$. This is *phase kickback*: the ancilla is unchanged and the sign lands on the input register.
**Why the other options are incorrect.**
**(A)** The ancilla does not store $f(x)$ here; that would be the case for an ancilla in $|0\rangle$.
**(B)** This drops the sign, which is the entire point of the construction.
**(D)** The input register is not overwritten; it remains $|x\rangle$.
**Key takeaway.** Ancilla in $|-\rangle$ turns the bit oracle into the phase oracle.

**Q.25** Single-correct MCQ · No kickback with $|+\rangle$

With $U_f$ as in the previous question and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, the state $U_f |x\rangle |+\rangle$ equals

**(A)** $|x\rangle |-\rangle$ **(B)** $(-1)^{f(x)} |x\rangle |-\rangle$ **(C)** $(-1)^{f(x)} |x\rangle |+\rangle$ **(D)** ✓ $|x\rangle |+\rangle$

**Correct answer: (D) Detailed solution**

**Concepts required.** $|+\rangle$ is an eigenvector of $X$ with eigenvalue $+1$, so a controlled bit flip leaves it unchanged.

**Relevant formulae.** $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$; $X |+\rangle = + |+\rangle$.

**Step-by-step explanation.** $U_f |x\rangle |+\rangle = \frac{1}{\sqrt{2}} |x\rangle \left( |f(x)\rangle + |1 \oplus f(x)\rangle \right) = |x\rangle |+\rangle$ in both cases, because $|0\rangle + |1\rangle$ is symmetric under the bit flip ($X |+\rangle = + |+\rangle$). No phase is kicked back, so the ancilla must be prepared in $|-\rangle$, not $|+\rangle$.

**Why the other options are incorrect.**

**(A)** $U_f$ cannot change the ancilla from $|+\rangle$ to $|-\rangle$: flipping $|0\rangle + |1\rangle$ gives the same state.

**(B)** This is the result for an ancilla starting in $|-\rangle$.

**(C)** The sign requires the eigenvalue $-1$, that is the ancilla $|-\rangle$.

**Key takeaway.** The ancilla's eigenvalue under $X$ decides the kicked-back sign.

**Q.26** Single-correct MCQ · $X$ conjugation selects the item

Let CCZ be the three-qubit gate that multiplies only $|111\rangle$ by $-1$. Which basis state has its sign flipped by $X_2 \, \mathrm{CCZ} \, X_2$, where $X_2$ is $X$ on qubit 2?

**(A)** $|111\rangle$ **(B)** $|010\rangle$ **(C)** ✓ $|101\rangle$ **(D)** $|000\rangle$

**Correct answer: (C) Detailed solution**

**Concepts required.** Conjugating by $X$ moves the affected basis state: $X_2 \, \mathrm{CCZ} \, X_2 = I - 2 \, |101\rangle\langle 101|$.

**Relevant formulae.** $\mathrm{CCZ} = I - 2 \, |111\rangle\langle 111|$; $U(I - 2 \, |a\rangle\langle a|)U^\dagger = I - 2U \, |a\rangle\langle a| \, U^\dagger$; $X_2$ flips qubit 2.

**Step-by-step explanation.** The rightmost $X_2$ acts first. A state picks up the sign exactly when $X_2$ maps it to $|111\rangle$, which happens for $|101\rangle$. The last $X_2$ restores the bit. So the sign flip sits on $|101\rangle$. This is the standard oracle recipe: apply $X$ wherever the marked string has a 0.

**Why the other options are incorrect.**

**(A)** $|111\rangle$ is the state for CCZ alone; after $X_2$ it is mapped to $|101\rangle$, which receives no sign.

**(B)** $|010\rangle$ is the bit-complement of the right answer.

**(D)** $|000\rangle$ would need $X$ on all three qubits.

**Key takeaway.** Put an $X$ on every qubit where the marked string has a 0.

**Q.27** Single-correct MCQ · CCZ from Toffoli

Let CCX be the Toffoli gate with target qubit 3. The product $(I \otimes I \otimes H)\,\mathrm{CCX}\,(I \otimes I \otimes H)$ equals

**(A)** CCX **(B)** $I \otimes I \otimes I$ **(C)** $Z \otimes Z \otimes Z$ **(D)** ✓ CCZ

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $HXH = Z$ and $HZH = X$. Conjugating the target of a controlled gate conjugates the gate itself.
**Relevant formulae.** $HXH = Z$, $HIH = I$; a controlled gate acts on its target only in the branch where all the controls are 1.
**Step-by-step explanation.** Use $HXH = Z$ and $HIH = I$. In the branch where both controls are 1, the target sees $HXH = Z$; in all other branches it sees $HIH = I$. So the product is the doubly controlled $Z$, i.e. $\mathrm{CCZ} = \mathrm{diag}(1, \ldots, 1, -1)$.
**Why the other options are incorrect.**
**(A)** CCX is the gate before conjugation; $HXH \neq X$.
**(B)** The product cannot be $I$ because in the branch $|11\rangle$ the target sees $Z$.
**(C)** $Z \otimes Z \otimes Z$ flips the sign for every odd-weight string, not only for $|111\rangle$.
**Key takeaway.** A multi-controlled $Z$ is a multi-controlled $X$ with Hadamards around the target.

**Q.28** Single-correct MCQ · Global vs. relative phase

Which pair of states are physically *indistinguishable* (they give the same statistics for every possible measurement)?

**(A)** ✓ $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $-\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
**(B)** $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$
**(C)** $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
**(D)** $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Global phase: $e^{i\gamma}|\psi\rangle$ is the same physical state. A relative phase is observable.
**Relevant formulae.** Same physical state: $|\psi'\rangle = e^{i\gamma}|\psi\rangle$ for a real $\gamma$. $|\langle\phi|\psi'\rangle|^2 = |\langle\phi|\psi\rangle|^2$ for all $|\phi\rangle$.
**Step-by-step explanation.** Two state vectors describe the same physical state iff they differ by one overall factor $e^{i\gamma}$. Only the first pair does: the factor is $-1$. In each of the other pairs the *relative* phase differs, and a suitable measurement separates them. For example $H$ maps $|+\rangle$ and $|-\rangle$ to $|0\rangle$ and $|1\rangle$. The minus sign in the diffusion circuit (see Q.47) is of the harmless global kind.
**Why the other options are incorrect.**
**(B)** These two Bell states are orthogonal, so a suitable measurement separates them.
**(C)** $|+\rangle$ and $|-\rangle$ are orthogonal, and an $X$-basis measurement separates them perfectly.
**(D)** These differ by a relative phase $i$ and have different $\langle Y\rangle$.
**Key takeaway.** Test with one common factor across all components: if none exists, the states differ.

**Q.29** Single-correct MCQ · Subspace measurement

Let $|\psi\rangle = \frac{1}{\sqrt{10}}(\,|00\rangle + 2\,|01\rangle - 2\,|10\rangle + i\,|11\rangle\,)$. A measurement asks only whether the two qubits have *different* values (outcome "different" on the span of $|01\rangle, |10\rangle$). The probability of "different" and the state afterwards are

**(A)** $\frac{4}{5}$ and $|01\rangle$

**(B)** ✓ $\frac{4}{5}$ and $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

**(C)** $\frac{4}{5}$ and $\frac{1}{\sqrt{10}}(2\,|01\rangle - 2\,|10\rangle)$

**(D)** $\frac{2}{5}$ and $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Projective measurement: sum the probabilities in the subspace, project, and re-normalise.

**Relevant formulae.** $P = \sum_{x\in S} |c_x|^2$ for the subspace spanned by the set $S$; the new state is $\frac{1}{\sqrt{P}}\sum_{x\in S} c_x\,|x\rangle$.

**Step-by-step explanation.** Sum the probabilities in the subspace: $P = \frac{4+4}{10} = \frac{4}{5}$. The post-measurement state is the projected vector, re-normalised: $\frac{(2|01\rangle - 2|10\rangle)/\sqrt{10}}{\sqrt{4/5}} = \frac{2|01\rangle - 2|10\rangle}{\sqrt{8}} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. Relative phases inside the subspace survive, and the measurement does not pick one of the basis states.

**Why the other options are incorrect.**

**(A)** The measurement does not distinguish $|01\rangle$ from $|10\rangle$, so it does not collapse to one of them.

**(C)** This vector was not re-normalised; its length is $\sqrt{4/5} \neq 1$.

**(D)** $\frac{2}{5}$ uses the amplitudes instead of squares or counts only one of the two terms.

**Key takeaway.** A coarse measurement keeps superpositions inside the measured subspace.

**Q.30** Single-correct MCQ · Unitary vs. projector

Let $|u\rangle$ be a unit vector in a space of dimension $N \geq 2$. Which of these operators is *not* unitary?

**(A)** $H^{\otimes n}$ (for $N = 2^n$)

**(B)** ✓ $|u\rangle\langle u|$

**(C)** $I - 2\,|u\rangle\langle u|$

**(D)** $2\,|u\rangle\langle u| - I$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Unitary means $U^\dagger U = I$, and hence invertible. Projectors have eigenvalue 0.

**Relevant formulae.** Unitary: $U^\dagger U = I$. Projector: $\Pi = |u\rangle\langle u|$, $\Pi^2 = \Pi$. Reflection: $R = R^\dagger$, $R^2 = I$.

**Step-by-step explanation.** $|u\rangle\langle u|$ is a projector: it has eigenvalue 0, so it is not invertible and cannot be unitary. A reflection $R = \pm(I - 2\,|u\rangle\langle u|)$ is Hermitian with $R^2 = I$, so $R^\dagger R = I$. $H^{\otimes n}$ is a tensor product of unitaries.

**Why the other options are incorrect.**

**(A)** A tensor product of unitaries is unitary.

**(C)** This is a reflection: Hermitian and squares to $I$, so it is unitary.

**(D)** This is also a reflection, the negative of the previous one.

**Key takeaway.** Projectors are not gates; reflections are.

**Q.31** Single-correct MCQ · Overlap $\cos\theta$

For $N = 16$ let $|s\rangle$ be the uniform superposition of all 16 basis states and $|r\rangle$ the uniform superposition of 15 of them (all except one). The value of $\langle r|s\rangle$ is

**(A)** ✓ $\frac{\sqrt{15}}{4}$ **(B)** $\frac{15}{16}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{1}{\sqrt{15}}$

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Overlap is the sum of products of matching amplitudes. Here $\cos\theta = \langle r|s\rangle$ and $\sin\theta = \langle m|s\rangle$.

**Relevant formulae.** $\langle r|s\rangle = \sum_x \overline{r_x} s_x$; with $|s\rangle$ uniform over $N$ states and $|r\rangle$ uniform over $N - M$: $\langle r|s\rangle = \sqrt{(N-M)/N} = \cos\theta$.

**Step-by-step explanation.** $\langle r|s\rangle = \sum_x \frac{1}{\sqrt{15}} \cdot \frac{1}{\sqrt{16}}$ over the 15 common basis states, which is $\frac{15}{\sqrt{15}\cdot 4} = \frac{\sqrt{15}}{4}$. It equals $\cos\theta$ with $\sin\theta = \frac{1}{4}$. $\frac{15}{16}$ is $\cos^2\theta$: the probability, not the overlap.

**Why the other options are incorrect.**

**(B)** $\frac{15}{16}$ is $|\langle r|s\rangle|^2$, the probability.

**(C)** $\frac{1}{4} = \sin\theta$ is the overlap of $|s\rangle$ with the *marked* state.

**(D)** $\frac{1}{\sqrt{15}}$ is a single amplitude of $|r\rangle$.

**Key takeaway.** The overlaps of $|s\rangle$ with the two axes define $\theta$.

**Q.32** Single-correct MCQ · State-vector memory

A state vector of $n = 33$ qubits is stored in double precision (16 bytes per complex amplitude). The memory needed is

**(A)** 256 GiB **(B)** 16 GiB **(C)** 64 GiB **(D)** ✓ 128 GiB

**Correct answer: (D)** **Detailed solution**

**Concepts required.** Memory $= 2^n\times$ bytes per amplitude, and $2^{30}$ bytes $= 1$ GiB.

**Relevant formulae.** Memory $= 2^n\times$ (bytes per amplitude); double precision uses 16 bytes; 1 GiB $= 2^{30}$ bytes.

**Step-by-step explanation.** $2^{33} \times 16 = 2^{33} \times 2^4 = 2^{37}$ bytes $= 2^7 \times 2^{30}$ bytes $= 128$ GiB. Every extra qubit doubles it. (16 GiB is the value for $n = 30$.)

**Why the other options are incorrect.**

**(A)** 256 GiB is the value for $n = 34$.

**(B)** 16 GiB is the value for $n = 30$.

**(C)** 64 GiB is the value for $n = 32$, or for $n = 33$ in single precision.

**Key takeaway.** Memory doubles with every qubit: $n = 30$ needs 16 GiB in double precision.

**Q.33** Single-correct MCQ · Diffusion for one qubit

For a single qubit ($n = 1$), the operator $2\,|+\rangle\langle+| - I$ equals

**(A)** ✓ $X$ **(B)** $H$ **(C)** $Z$ **(D)** $-I$

**Correct answer: (A)** **Detailed solution**

**Concepts required.** $|+\rangle\langle+|$ has all entries $\frac{1}{2}$; the reflection about $|+\rangle$ fixes $|+\rangle$ and negates $|-\rangle$.
**Relevant formulae.** $|+\rangle\langle+| = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$; $X = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$.
**Step-by-step explanation.** $|+\rangle\langle+| = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, so $2\,|+\rangle\langle+| - I = \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right) - \left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right) = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right) = X$. *Check:* $X\,|+\rangle = |+\rangle$ and $X\,|-\rangle = -\,|-\rangle$, which are the eigenvalues required of this reflection.
**Why the other options are incorrect.**

**(B)** $H$ does not fix $|+\rangle$: it maps $|+\rangle$ to $|0\rangle$.

**(C)** $Z$ fixes $|0\rangle$ and negates $|1\rangle$: that is the reflection about $|0\rangle$.

**(D)** $-I$ would negate $|+\rangle$ too, not fix it.

**Key takeaway.** Diffusion on one qubit is simply $X$.

**Q.34** Single-correct MCQ · $|s\rangle$ is fixed by $D$

For $n$ qubits let $|s\rangle = H^{\otimes n}\,|0\ldots0\rangle$ and $D = 2\,|s\rangle\langle s| - I$. If the oracle marks nothing (it acts as the identity), then after $k$ iterations the state is

**(A)** $(-1)^k\,|s\rangle$ **(B)** $|0\ldots0\rangle$ for odd $k$
**(C)** a state whose amplitudes depend on $k$ **(D)** ✓ $|s\rangle$ for every $k$

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $|s\rangle$ is the $+1$ eigenvector of $D$.
**Relevant formulae.** $D\,|s\rangle = 2\,|s\rangle\,\langle s|s\rangle - |s\rangle = |s\rangle$, with $\langle s|s\rangle = 1$.
**Step-by-step explanation.** $D\,|s\rangle = 2\,|s\rangle\,\langle s|s\rangle - |s\rangle = |s\rangle$, so $|s\rangle$ is an eigenvector of $D$ with eigenvalue $+1$. With $U_\omega = I$, each iteration is just $D$ and leaves $|s\rangle$ unchanged. Measuring gives a uniformly random string, which matches $\sin^2\theta = 0$: nothing can be amplified when no item is marked.
**Why the other options are incorrect.**

**(A)** The eigenvalue is $+1$, so no factor $(-1)^k$ arises. (The circuit form $-D$ would give such a factor, but it is a global phase.)

**(B)** $D$ does not map $|s\rangle$ to $|0\ldots0\rangle$; that is what $H^{\otimes n}$ does.

**(C)** The amplitudes stay at $1/\sqrt{N}$ because $|s\rangle$ is an eigenvector.

**Key takeaway.** Without a marked item the algorithm does nothing, as $\theta = 0$ predicts.

**Q.35** Matching lists · Single-qubit gate actions

Match each expression in List-I with its value in List-II.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | $H\,\|1\rangle$ | **(1)** | $-\,\|-\rangle$ |
| **(Q)** | $Z\,\|-\rangle$ | **(2)** | $\|0\rangle$ |
| **(R)** | $X\,\|-\rangle$ | **(3)** | $\|+\rangle$ |
| **(S)** | $H\,\|+\rangle$ | **(4)** | $\|-\rangle$ |

**(A)** P→4, Q→1, R→3, S→2 **(B)** P→4, Q→3, R→2, S→1
**(C)** ✓ P→4, Q→3, R→1, S→2 **(D)** P→3, Q→4, R→1, S→2

**Correct answer: (C)** **Detailed solution**

**Concepts required.** $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, $Z$ exchanges $|\pm\rangle$, and $X|\pm\rangle = \pm|\pm\rangle$.
**Relevant formulae.** $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, $H^2 = I$; $Z|\pm\rangle = |\mp\rangle$; $X|\pm\rangle = \pm|\pm\rangle$.
**Step-by-step explanation.** $H|1\rangle = |-\rangle$ (4). $Z|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$ (3). $X|-\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -|-\rangle$ (1). $H|+\rangle = |0\rangle$ because $H^2 = I$ (2). Hence P→4, Q→3, R→1, S→2.
**Why the other options are incorrect.**
**(A)** This gives $Z|-\rangle = -|-\rangle$, which is the action of $X$; $Z$ exchanges $|\pm\rangle$.
**(B)** This exchanges R and S: $X|-\rangle$ cannot be $|0\rangle$, and $H|+\rangle$ is $|0\rangle$.
**(D)** This exchanges P and Q: $H|1\rangle$ is $|-\rangle$, and $Z|-\rangle$ is $|+\rangle$.
**Key takeaway.** Keep the table of $H, X, Z$ on $|0\rangle, |1\rangle, |\pm\rangle$ at your fingertips.

**Q.36** MATCHING LISTS · Product vs. entangled states

Match each two-qubit state in List-I with its description in List-II.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | $\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle + \|11\rangle)$ | **(1)** | entangled, with four amplitudes of equal size |
| **(Q)** | $\frac{1}{2}(\|00\rangle - \|01\rangle - \|10\rangle + \|11\rangle)$ | **(2)** | the product $\|-\rangle\|-\rangle$ |
| **(R)** | $\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle - \|11\rangle)$ | **(3)** | entangled, with only two non-zero amplitudes |
| **(S)** | $\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$ | **(4)** | the product $\|+\rangle\|+\rangle$ |

**(A)** P→4, Q→2, R→3, S→1
**(B)** ✓ P→4, Q→2, R→1, S→3
**(C)** P→4, Q→1, R→2, S→3
**(D)** P→2, Q→4, R→1, S→3

**Correct answer: (B)** **Detailed solution**

**Concepts required.** A two-qubit state is a product iff $c_{00}c_{11} = c_{01}c_{10}$.
**Relevant formulae.** $\sum_{j,k} c_{jk}|jk\rangle$ is a product state $\iff c_{00}c_{11} = c_{01}c_{10}$.
**Step-by-step explanation.** Test $c_{00}c_{11}$ against $c_{01}c_{10}$; they are equal exactly for a product state. P: $\frac{1}{4} = \frac{1}{4}$, product, equal to $|+\rangle|+\rangle$ (4). Q: $\frac{1}{4} = \frac{1}{4}$, product $|-\rangle|-\rangle$ (2). R: $-\frac{1}{4} \neq \frac{1}{4}$, entangled with four equal magnitudes (1); this is the state after the oracle for $n = 2$. S: $\frac{1}{2} \neq 0$, entangled with two non-zero amplitudes (3).
**Why the other options are incorrect.**
**(A)** This exchanges R and S: state R has four non-zero amplitudes, and state S has two.
**(C)** This calls Q entangled, but it passes the product test and equals $|-\rangle|-\rangle$.
**(D)** This exchanges P and Q: all plus signs give $|+\rangle|+\rangle$.
**Key takeaway.** One sign flip on $|s\rangle$ with $n = 2$ gives an entangled state, while Hadamards alone never do.

**Q.37** Assertion–Reason · Why the oracle alone reveals nothing

**Assertion (A):** Measuring in the computational basis immediately after a phase oracle acts on $|s\rangle$ gives a uniformly random string; the oracle step alone reveals nothing.

**Reason (R):** The oracle multiplies each amplitude by $\pm 1$, and the Born probabilities depend only on the magnitudes of the amplitudes.

**(A)** ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

**(B)** Both A and R are true, but R is *not* the correct explanation of A

**(C)** A is true, but R is false

**(D)** A is false, but R is true

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Probabilities depend only on $|c_x|^2$, while the oracle changes only signs.
**Relevant formulae.** $U_\omega |x\rangle = (-1)^{f(x)} |x\rangle$; Born rule $P(x) = |c_x|^2$; $|\pm a|^2 = |a|^2$.
**Step-by-step explanation.** **R is true:** $U_\omega |x\rangle = (-1)^{f(x)} |x\rangle$ and $|\pm a|^2 = |a|^2$. **A is true** because of R: the magnitudes stay $1/\sqrt{N}$, so the distribution stays uniform. R is the correct explanation. Answer: (A).
**Why the other options are incorrect.**

**(B)** R does explain A: it gives the exact reason that the distribution stays uniform.

**(C)** R is true: $|\pm a|^2 = |a|^2$.

**(D)** A is true: the output is uniformly random whichever item is marked.

**Key takeaway.** The oracle's effect is invisible until an interfering step such as diffusion follows.

**Q.38** Statement I / II · Relative phases are observable

**Statement I:** The states $|\psi\rangle$ and $-|\psi\rangle$ give identical statistics for every measurement.

**Statement II:** Multiplying only *one* basis-state amplitude of a superposition by $-1$ can never change the statistics of any measurement.

**(A)** Both Statement I and Statement II are true

**(B)** Both Statement I and Statement II are false

**(C)** ✓ Statement I is true, but Statement II is false

**(D)** Statement I is false, but Statement II is true

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Global phases are unobservable; relative phases can be observed after interference.
**Relevant formulae.** Global phase: $e^{i\gamma} |\psi\rangle$. Relative phase: a phase on only some terms, as in $|0\rangle - |1\rangle$ against $|0\rangle + |1\rangle$.
**Step-by-step explanation.** **I is true:** a global phase cancels in $|\langle\phi|\psi\rangle|^2$ for every $|\phi\rangle$. **II is false:** that is a *relative* phase. Counterexample: $|+\rangle$ and $|-\rangle$ differ by the sign of $|1\rangle$ only, yet a measurement in the $\{|+\rangle, |-\rangle\}$ basis separates them perfectly. Grover's algorithm exists because the sign is detectable after interference. Answer: (C).
**Why the other options are incorrect.**

**(A)** II is false: $|+\rangle$ and $|-\rangle$ differ only by the sign of one amplitude and yet are distinguishable.

**(B)** I is true: a global phase cancels in every probability.

**(D)** I is true, so this option fails.

**Key takeaway.** A sign on one amplitude is a relative phase: it is exactly the signal that Grover amplifies.

**Paragraph for Questions 39 and 40**

Three qubits start in $|0\rangle\,|0\rangle\,|1\rangle$ (input qubits 1 and 2, ancilla qubit 3). A Hadamard gate is applied to all three. Then the Toffoli gate is applied with qubits 1 and 2 as controls and qubit 3 as target: it implements $U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle$ with $f(x_1x_2) = x_1 \wedge x_2$.

**Q.39** PARAGRAPH BASED · Toffoli as a phase oracle

The state of the three qubits after the Toffoli gate is

**(A)** $\frac{1}{2}(\,|00\rangle - |01\rangle - |10\rangle - |11\rangle\,)\,|-\rangle$

**(B)** $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)\,|+\rangle$

**(C)** $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle + |11\rangle\,)\,|-\rangle$

**(D)** ✓ $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)\,|-\rangle$

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $H\,|1\rangle = |-\rangle$; kickback puts the sign $(-1)^{f(x)}$ on $|x\rangle$.
**Relevant formulae.** $H\,|1\rangle = |-\rangle$; $U_f\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle$; here $f(x_1x_2) = x_1 \wedge x_2$.
**Step-by-step explanation.** After the Hadamards: $\frac{1}{2}\sum_x |x\rangle\,|-\rangle$, since $H\,|1\rangle = |-\rangle$. By kickback each term gets $(-1)^{f(x)}$, and $f = 1$ only for $x = 11$. So the state is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)\,|-\rangle$. The ancilla stays in $|-\rangle$, unentangled with the inputs.
**Why the other options are incorrect.**

**(A)** Only $x = 11$ has $f = 1$, so only one sign flips.

**(B)** The ancilla remains in $|-\rangle$; it does not change to $|+\rangle$.

**(C)** This forgets the kicked-back sign on $|11\rangle$.

**Key takeaway.** Toffoli plus ancilla $|-\rangle$ is the phase oracle for $x_1 \wedge x_2$.

**Q.40** PARAGRAPH BASED · Ancilla stays unentangled

Now the ancilla alone is measured in the computational basis. The probability of the outcome 1, and the state of the input qubits afterwards, are

**(A)** $\frac{1}{4}$; the input qubits collapse to $|11\rangle$

**(B)** ✓ $\frac{1}{2}$; the input qubits remain in $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)$

**(C)** $\frac{1}{2}$; the input qubits collapse to $|11\rangle$

**(D)** 1; the input qubits remain in $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** A measurement on one factor of a product state does not disturb the other factor.
**Relevant formulae.** For a product state $|\phi\rangle\,|-\rangle$, measuring the second factor leaves $|\phi\rangle$ unchanged; $P(1) = |\,\langle 1|-\rangle\,|^2 = \frac{1}{2}$.
**Step-by-step explanation.** The state after the gate is a product of the input state and $|-\rangle$ (see Q.39). Measuring $|-\rangle$ gives 1 with probability $\frac{1}{2}$, and because the ancilla is not entangled with the inputs, the input qubits are not disturbed.
**Why the other options are incorrect.**

**(A)** This treats the ancilla as if it were in $|f(x)\rangle$, i.e. started in $|0\rangle$.

**(C)** The probability is right, but the inputs do not collapse because there is no entanglement.

**(D)** $|-\rangle$ gives outcome 1 with probability $\frac{1}{2}$, not 1.

**Key takeaway.** With an ancilla in $|-\rangle$ the ancilla stays unentangled and can be re-used.

## SECTION C · Grover's Algorithm Fundamentals Q.41–60

**Q.41** Single-correct MCQ · $N = 4$ is exact in one step

Grover's algorithm is run on $n = 2$ qubits ($N = 4$) with one marked item. After exactly one iteration, the probability of measuring the marked item is

**(A)** $\frac{1}{2}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{3}{4}$ **(D)** ✓ 1

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $\sin\theta = 1/\sqrt{N}$ and $P_k = \sin^2((2k+1)\theta)$. For $N = 4$, $\theta = 30°$.
**Relevant formulae.** $\sin\theta = \sqrt{M/N}$, with $N$ items and $M$ marked; $P_k = \sin^2((2k+1)\theta)$ after $k$ iterations.
**Step-by-step explanation.** $\sin\theta = \frac{1}{\sqrt{4}} = \frac{1}{2}$, so $\theta = 30°$. After $k = 1$ iteration $P = \sin^2(3\theta) = \sin^2 90° = 1$. In amplitudes: after the oracle the entries are $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ with mean $\frac{1}{4}$; inversion about the mean gives $(0, 0, 0, 1)$.
**Why the other options are incorrect.**
**(A)** $\frac{1}{2}$ is the result for $M = N/2$, not for this case.
**(B)** $\frac{1}{4}$ is the probability before any iteration.
**(C)** $\frac{3}{4}$ is the total probability of the unmarked items at the start.
**Key takeaway.** $N = 4$ is the unique single-item case where one iteration is exact.

**Q.42** Single-correct MCQ · One iteration, $N = 8$

For $N = 8$ with one marked item, the probability of measuring the marked item after one iteration is

**(A)** 1 **(B)** ✓ $\frac{25}{32}$ **(C)** $\frac{5}{8}$ **(D)** $\frac{9}{16}$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Trace the amplitudes with inversion about the mean, or use $P_1 = s(3 - 4s)^2$ with $s = 1/N$.
**Relevant formulae.** Oracle: marked $c \mapsto -c$. Diffusion: $c_x \mapsto 2\bar{c} - c_x$ with $\bar{c} = \frac{1}{N}\sum_x c_x$. Also $P_1 = s(3 - 4s)^2$, with $s = M/N$.
**Step-by-step explanation.** Write $a = 1/\sqrt{8}$. After the oracle: seven entries $a$, one entry $-a$. Mean $= \frac{6a}{8} = \frac{3}{4}a$. The marked entry becomes $2 \cdot \frac{3}{4}a + a = \frac{5}{2}a$, so $P = \frac{25}{4}a^2 = \frac{25}{4} \cdot \frac{1}{8} = \frac{25}{32}$. *Check:* $\sin^2 3\theta = s(3 - 4s)^2$ with $s = \frac{1}{8}$ gives $\frac{1}{8} \cdot \frac{25}{4} = \frac{25}{32}$.
**Why the other options are incorrect.**
**(A)** $P = 1$ happens for $N = 4$ only; here $3\theta \neq 90°$.
**(C)** $\frac{5}{8}$ uses the amplitude factor $\frac{5}{2}$ divided by 4 and forgets to square.
**(D)** $\frac{9}{16}$ uses the mean $\frac{3}{4}$ squared instead of the new amplitude.
**Key takeaway.** For small $N$, inversion about the mean is the fastest method.

**Q.43** SINGLE-CORRECT MCQ · $k_{\text{opt}}$ and $P$ for $n = 4$

For $n = 4$ qubits with one marked item ($\theta \approx 0.2527$ rad), the optimal number of iterations $k_{\text{opt}}$ and the resulting success probability are

**(A)** $k_{\text{opt}} = 2$ and $P \approx 0.908$
**(B)** $k_{\text{opt}} = 4$ and $P \approx 0.582$
**(C)** $k_{\text{opt}} = 3$ and $P = 1$
**(D)** ✓ $k_{\text{opt}} = 3$ and $P \approx 0.961$

**Correct answer: (D) Detailed solution**

**Concepts required.** $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$ and $P = \sin^2((2k+1)\theta)$.
**Relevant formulae.** $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$; $P_k = \sin^2((2k+1)\theta)$; $\sin\theta = 1/\sqrt{N}$ for one marked item.
**Step-by-step explanation.** $\frac{\pi}{4\theta} - \frac{1}{2} = \frac{0.7854}{0.2527} - 0.5 = 3.108 - 0.5 = 2.608$, which rounds to $k_{\text{opt}} = 3$. Then $P = \sin^2(7\theta) = \sin^2(1.769) \approx 0.961$. The angle $7\theta$ does not hit $\frac{\pi}{2}$ exactly, so $P < 1$. For comparison, $k = 2$ gives $\sin^2 5\theta \approx 0.908$ and $k = 4$ gives $\sin^2 9\theta \approx 0.582$.
**Why the other options are incorrect.**
**(A)** $k = 2$ is below the optimum: $5\theta$ is further from $\frac{\pi}{2}$ than $7\theta$.
**(B)** $k = 4$ over-rotates, and $P$ falls to 0.58.
**(C)** $k = 3$ is right but $P < 1$: $7\theta = 1.769$ rad, not $\frac{\pi}{2}$.
**Key takeaway.** Optimal does not mean perfect: $P$ is close to 1 but usually not equal to 1.

**Q.44** SINGLE-CORRECT MCQ · Matrix of $D$ for $N = 4$

For $n = 2$ ($N = 4$), the matrix of the diffusion operator $D = 2|s\rangle\langle s| - I$ has

**(A)** diagonal entries $+\frac{1}{2}$ and off-diagonal entries $-\frac{1}{2}$
**(B)** ✓ diagonal entries $-\frac{1}{2}$ and off-diagonal entries $+\frac{1}{2}$
**(C)** diagonal entries $-1$ and off-diagonal entries $+1$
**(D)** all entries equal to $\frac{1}{2}$

**Correct answer: (B) Detailed solution**

**Concepts required.** $|s\rangle\langle s|$ has every entry $\frac{1}{N}$, so $D$ has diagonal $\frac{2}{N} - 1$ and off-diagonal $\frac{2}{N}$.
**Relevant formulae.** $D = 2|s\rangle\langle s| - I$; $(|s\rangle\langle s|)_{xy} = \frac{1}{N}$; so $D_{xx} = \frac{2}{N} - 1$ and $D_{xy} = \frac{2}{N}$ for $x \neq y$.
**Step-by-step explanation.** Every entry of $|s\rangle\langle s|$ is $\frac{1}{\sqrt{4}} \cdot \frac{1}{\sqrt{4}} = \frac{1}{4}$, so every entry of $2|s\rangle\langle s|$ is $\frac{1}{2}$. Subtracting $I$ changes only the diagonal: $\frac{1}{2} - 1 = -\frac{1}{2}$. In general the diagonal is $\frac{2}{N} - 1$ and the off-diagonal is $\frac{2}{N}$. *Check:* acting on the amplitudes $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$, the last row gives $\frac{1}{2}(\frac{1}{2}+\frac{1}{2}+\frac{1}{2}) - \frac{1}{2}(-\frac{1}{2}) = \frac{3}{4} + \frac{1}{4} = 1$, the same as inversion about the mean.
**Why the other options are incorrect.**
**(A)** This is $-D = I - 2|s\rangle\langle s|$, which is what the circuit implements; it differs from $D$ by a global phase.
**(C)** Forgets the factor $\frac{1}{4}$ from the normalisation of $|s\rangle$; this matrix is not unitary.
**(D)** This is $2|s\rangle\langle s|$, before $I$ has been subtracted.
**Key takeaway.** Row by row, $D$ computes $2\bar{c} - c_x$, which is inversion about the mean.

**Q.45** Single-correct MCQ · Building the oracle

An oracle for the marked string $|0101\rangle$ is built as: $X$ gates, then the multi-controlled $Z$ on all four qubits, then the same $X$ gates. The $X$ gates act on

**(A)** qubits 2 and 4 **(B)** no qubit **(C)** ✓ qubits 1 and 3 **(D)** all four qubits

**Correct answer: (C)** **Detailed solution**

**Concepts required.** The multi-controlled $Z$ flips $|1\dots1\rangle$ only, so the $X$ gates must map the marked string to all ones.
**Relevant formulae.** $\mathrm{C}^{n-1}\mathrm{Z} = I - 2\,|1\dots1\rangle\langle1\dots1|$; $X\,|0\rangle = |1\rangle$; so $X$ on the 0 bits of $x^*$ maps $|x^*\rangle$ to $|1\dots1\rangle$.
**Step-by-step explanation.** The multi-controlled $Z$ flips the sign of $|1111\rangle$ only, so the $X$ layer must map the marked string to $|1111\rangle$. That means flipping exactly the bits where $x^*$ has a 0. In 0101 these are qubits 1 and 3. The second $X$ layer undoes the first, so the sign flip ends up on $|0101\rangle$.
**Why the other options are incorrect.**
**(A)** Qubits 2 and 4 hold 1 already: flipping them maps $|1010\rangle$ to all ones, the wrong item.
**(B)** With no $X$ gates the oracle marks $|1111\rangle$.
**(D)** Flipping all qubits marks the complement $|1010\rangle$.
**Key takeaway.** Rule: $X$ on the zeros of $x^*$.

**Q.46** Single-correct MCQ · Gate count per iteration

Use the circuit in the conventions box and count each multi-controlled $Z$ as one gate. The number of gates in *one* Grover iteration for $n = 6$, $x^* = 010011$ is

**(A)** ✓ 32 **(B)** 30 **(C)** 26 **(D)** 38

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Per iteration: oracle $2z + 1$ gates and diffusion $4n + 1$ gates, so $4n + 2z + 2$ in total.
**Relevant formulae.** Gates per iteration $= (2z + 1) + (4n + 1) = 4n + 2z + 2$, with $n$ qubits and $z$ zeros in $x^*$.
**Step-by-step explanation.** The oracle has $2z$ $X$ gates, where $z$ is the number of 0s in $x^*$ (here $z = 3$), and one multi-controlled $Z$: $6 + 1 = 7$ gates. The diffusion has $2n$ $H$, $2n$ $X$ and one multi-controlled $Z$: $12 + 12 + 1 = 25$ gates. Total $= 4n + 2z + 2 = 24 + 6 + 2 = 32$.
**Why the other options are incorrect.**
**(B)** 30 omits the two multi-controlled $Z$ gates.
**(C)** 26 omits the $2z$ oracle $X$ gates.
**(D)** 38 counts $2n$ oracle $X$ gates, as if every bit were 0.
**Key takeaway.** The formula $4n + 2z + 2$ is the one the notebook uses in its time estimate.

**Q.47** Single-correct MCQ · Diffusion circuit $= -D$

Let $C$ be the multi-controlled $Z$ on $n$ qubits. The circuit $H^{\otimes n} X^{\otimes n}\, C\, X^{\otimes n} H^{\otimes n}$ implements

**(A)** $2\,|s\rangle\langle s| - I = D$ **(B)** ✓ $I - 2\,|s\rangle\langle s| = -D$

**(C)** $I - 2\,|0\ldots0\rangle\langle 0\ldots0|$ **(D)** $I - 2\,|1\ldots1\rangle\langle 1\ldots1|$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** $U(I - 2\,|a\rangle\langle a|)U^\dagger = I - 2U\,|a\rangle\langle a|\,U^\dagger$; a reflection conjugated by a unitary is a reflection about the image.

**Relevant formulae.** $\mathrm{C}^{n-1}\mathrm{Z} = I - 2\,|1\ldots1\rangle\langle 1\ldots1|$; $U(I - 2\,|a\rangle\langle a|)U^\dagger = I - 2U\,|a\rangle\langle a|\,U^\dagger$; $D = 2\,|s\rangle\langle s| - I$.

**Step-by-step explanation.** $C = I - 2\,|1\ldots1\rangle\langle 1\ldots1|$. Conjugating by $X^{\otimes n}$ gives $I - 2\,|0\ldots0\rangle\langle 0\ldots0|$, and conjugating by $H^{\otimes n}$ gives $I - 2\,|s\rangle\langle s|$, since $H^{\otimes n}\,|0\ldots0\rangle = |s\rangle$. That is $-D$: the circuit equals the diffusion up to the global phase $-1$, which has no observable effect.

**Why the other options are incorrect.**

**(A)** This is the intended operator, but the circuit gives its negative. Only a global phase separates them, but the exact identity is $-D$.

**(C)** This is the operator at an intermediate step, before the $H$ layers are included.

**(D)** This is the multi-controlled $Z$ itself.

**Key takeaway.** Diffusion circuit $= -D$, and the minus sign is a harmless global phase.

**Q.48** Single-correct MCQ · Unmarked amplitude

For $N = 16$ with one marked item, the amplitude of each *unmarked* item after one iteration is

**(A)** $\frac{1}{4}$ **(B)** $\frac{11}{16}$ **(C)** ✓ $\frac{3}{16}$ **(D)** $\frac{7}{32}$

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Inversion about the mean: new amplitude $= 2 \times$ mean $-$ old, with the mean taken after the oracle.

**Relevant formulae.** $\bar{c} = \frac{1}{N}\sum_x c_x$ after the oracle; new amplitude $= 2\bar{c} - c_x$.

**Step-by-step explanation.** Start with amplitudes $\frac{1}{4}$. After the oracle the mean is $\frac{15-1}{16}\cdot\frac{1}{4} = \frac{7}{32}$. An unmarked entry becomes $2\cdot\frac{7}{32} - \frac{1}{4} = \frac{7}{16} - \frac{4}{16} = \frac{3}{16}$, and the marked entry becomes $\frac{7}{16} + \frac{4}{16} = \frac{11}{16}$. *Check:* $15\cdot\frac{9}{256} + \frac{121}{256} = 1$.

**Why the other options are incorrect.**

**(A)** $\frac{1}{4}$ is the initial amplitude.

**(B)** $\frac{11}{16}$ is the amplitude of the marked item.

**(D)** $\frac{7}{32}$ is the mean, not the new amplitude.

**Key takeaway.** Unmarked amplitudes shrink a little and the marked one grows a lot.

**Q.49** Single-correct MCQ · $M = N/4$

Let $N = 16$ with $M = 4$ marked items. After one iteration the probability of measuring a marked item is

**(A)** $\frac{3}{4}$ **(B)** $\frac{25}{32}$ **(C)** ✓ 1 **(D)** $\frac{1}{4}$

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Several solutions: $\sin^2\theta = M/N$. $M = N/4$ gives $\theta = 30°$.
**Relevant formulae.** $\sin^2\theta = M/N$; $P_1 = \sin^2 3\theta$.
**Step-by-step explanation.** $\sin^2\theta = \frac{M}{N} = \frac{1}{4}$, so $\theta = 30°$ and $P_1 = \sin^2 3\theta = \sin^2 90° = 1$. With $M = N/4$ a single iteration succeeds with certainty, for every $N$ for which $M = N/4$ is an integer. The final state is the uniform superposition of the four marked items.
**Why the other options are incorrect.**
**(A)** $\frac{3}{4}$ is the probability of the unmarked set before any iteration.
**(B)** $\frac{25}{32}$ is the value for $s = \frac{1}{8}$ and not for $s = \frac{1}{4}$.
**(D)** $\frac{1}{4}$ is the probability before any iteration.
**Key takeaway.** $M = N/4$ is the special ratio for a one-step exact search.

**Q.50** Single-correct MCQ · $M = N/2$

Let $N = 8$ with $M = 4$ marked items. The probability of a marked outcome after $k$ iterations is

**(A)** 1 for even $k$ and 0 for odd $k$ **(B)** 0 for $k = 1$
**(C)** 1 for $k = 1$ **(D)** ✓ $\frac{1}{2}$ for every $k$

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $M = N/2$ gives $\theta = 45°$, and $2\theta = 90°$ steps skip between the diagonals.
**Relevant formulae.** $\sin^2\theta = M/N$; $P_k = \sin^2((2k+1)\theta)$; $\sin^2 45° = \sin^2 135° = \frac{1}{2}$.
**Step-by-step explanation.** $\sin^2\theta = \frac{1}{2}$ gives $\theta = 45°$, and $P_k = \sin^2\left((2k+1)45°\right)$. The angles $45°, 135°, 225°, \ldots$ all have $\sin^2 = \frac{1}{2}$. The iteration rotates by $2\theta = 90°$, which exchanges the two diagonals and never brings the state closer to the marked axis. Grover gives no gain at $M = N/2$.
**Why the other options are incorrect.**
**(A)** The angles $45°, 135°, \ldots$ all have $\sin^2 = \frac{1}{2}$, so there is no oscillation between 0 and 1.
**(B)** That would need $3\theta = 180°$, i.e. $\theta = 60°$.
**(C)** That would need $3\theta = 90°$, i.e. $\theta = 30°$.
**Key takeaway.** At $M = N/2$, classical guessing is already as good as Grover.

**Q.51** Single-correct MCQ · $M = 3N/4$

Let $N = 8$ with $M = 6$ marked items. The probability of a marked outcome after one iteration is

**(A)** $\frac{3}{4}$ **(B)** $\frac{1}{4}$ **(C)** 1 **(D)** ✓ 0

**Correct answer: (D)** **Detailed solution**

**Concepts required.** $M > N/2$ gives $\theta > 45°$, and iterations can rotate past the marked axis at once.
**Relevant formulae.** $\sin^2\theta = M/N$; $P_1 = \sin^2 3\theta$; $\sin 180° = 0$.
**Step-by-step explanation.** $\sin^2\theta = \frac{3}{4}$ gives $\theta = 60°$, so $P_1 = \sin^2 180° = 0$. Amplitudes confirm this: with $a = 1/\sqrt{8}$ the oracle gives six entries $-a$ and two entries $+a$; the mean is $-\frac{1}{2}a$; the marked

entries become $-a + a = 0$. Measuring *without* any iteration ($P_0 = \frac{3}{4}$) is far better.

**Why the other options are incorrect.**

**(A)** $\frac{3}{4}$ is $P_0$, the probability with no iteration.

**(B)** $\frac{1}{4}$ is the initial probability of the unmarked items.

**(C)** 1 would need $3\theta = 90°$; here $3\theta = 180°$.

**Key takeaway.** If most items are marked, do not iterate: just measure.

**Q.52** SINGLE-CORRECT MCQ · Over-rotation at $2k_{\text{opt}}$

On $2^{15}$ items with one marked item, $k_{\text{opt}} = 142$. If $k = 284$ iterations are used instead, the probability of the marked item is about

**(A)** ✓ $3 \times 10^{-6}$ **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{32768}$ **(D)** 1

**Correct answer: (A)** **Detailed solution**

**Concepts required.** $P_k$ is periodic: at $k \approx 2k_{\text{opt}}$ the state has rotated by nearly $\pi$ and returns to near the unmarked axis.

**Relevant formulae.** $P_k = \sin^2((2k+1)\theta)$; $\sin^2(\pi + \delta) \approx \delta^2$ for a small $\delta$.

**Step-by-step explanation.** The angle is $(2k+1)\theta = 569 \times 0.0055243 = 3.1433$ rad, which is 0.0017 past $\pi$. So $P = \sin^2(569\theta) \approx (0.0017)^2 \approx 3 \times 10^{-6}$. This is over-rotation: twice the optimal number of iterations brings the state back near the unmarked axis, and the marked item is then *less* likely than a random guess ($1/32768 \approx 3 \times 10^{-5}$).

**Why the other options are incorrect.**

**(B)** $\frac{1}{2}$ is the value around $k \approx k_{\text{opt}}/2$, or $k \approx \frac{3}{2}k_{\text{opt}}$.

**(C)** $\frac{1}{32768} \approx 3 \times 10^{-5}$ is the starting value at $k = 0$, about ten times larger than the value at $k = 284$.

**(D)** 1 would be right near $k_{\text{opt}}$, not near $2k_{\text{opt}}$.

**Key takeaway.** More iterations are not always better: Grover is a rotation, not a monotonic filter.

**Q.53** SINGLE-CORRECT MCQ · Size of the speed-up

For $n = 20$ qubits with one marked item, Grover needs 804 iterations. A classical random search needs on average about $2^{19}$ queries. The classical cost divided by the quantum cost is about

**(A)** ✓ 650 **(B)** 2 **(C)** 1000 **(D)** 20

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Ratio $= \frac{N/2}{(\pi/4)\sqrt{N}} = \frac{2}{\pi}\sqrt{N}$.

**Relevant formulae.** Classical average $\approx N/2$; Grover $k \approx \frac{\pi}{4}\sqrt{N}$; ratio $= \frac{N/2}{(\pi/4)\sqrt{N}} = \frac{2}{\pi}\sqrt{N}$.

**Step-by-step explanation.** $2^{19} = 524\,288$ and $524\,288/804 \approx 652$. In general the ratio is $\frac{N/2}{(\pi/4)\sqrt{N}} = \frac{2}{\pi}\sqrt{N}$, and here $\frac{2}{\pi} \times 1024 \approx 652$. The speed-up is quadratic, not exponential.

**Why the other options are incorrect.**

**(B)** 2 would be a constant speed-up.

**(C)** 1000 rounds $\sqrt{N} = 1024$ and forgets the factor $\frac{2}{\pi}$.

**(D)** 20 is $n$; the ratio is not polynomial in $n$.

**Key takeaway.** Quadratic speed-up: the gain is $\frac{2}{\pi}\sqrt{N}$.

**Q.54** Single-correct MCQ · Shot statistics

A Grover circuit with success probability $P = 0.9613$ is run for 2000 shots. The expected marked count and its binomial standard deviation are about

**(A)** $1923 \pm 44$ **(B)** ✓ $1923 \pm 9$ **(C)** $2000 \pm 0$ **(D)** $1923 \pm 77$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Binomial: mean $SP$, standard deviation $\sqrt{SP(1-P)}$.
**Relevant formulae.** Binomial counts from $S$ shots: mean $= SP$, standard deviation $= \sqrt{SP(1-P)}$.
**Step-by-step explanation.** Mean $= SP = 2000 \times 0.9613 \approx 1923$. Standard deviation $= \sqrt{SP(1-P)} = \sqrt{2000 \times 0.9613 \times 0.0387} = \sqrt{74.4} \approx 8.6$. $\pm 44 = \sqrt{1923}$ is the Poisson estimate, which is wrong here because $P$ is close to 1.
**Why the other options are incorrect.**

**(A)** $\pm 44 = \sqrt{1923}$ uses a Poisson estimate and ignores the factor $1 - P$.

**(C)** $P < 1$, so not every shot succeeds.

**(D)** $\pm 77 = S(1-P)$ is the expected number of failures, not a spread.

**Key takeaway.** When $P$ is close to 1, the spread is small: compare measurement against $\sqrt{SP(1-P)}$.

**Q.55** Matching lists · $k_{\text{opt}}$ against $n$

Match the number of qubits (one marked item) in List-I with $k_{\text{opt}}$ in List-II. Use $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | $n = 2$ | **(1)** | 12 |
| **(Q)** | $n = 6$ | **(2)** | 25 |
| **(R)** | $n = 8$ | **(3)** | 1 |
| **(S)** | $n = 10$ | **(4)** | 6 |

**(A)** ✓ P→3, Q→4, R→1, S→2 **(B)** P→3, Q→1, R→4, S→2
**(C)** P→3, Q→4, R→2, S→1 **(D)** P→4, Q→3, R→1, S→2

**Correct answer: (A)** **Detailed solution**

**Concepts required.** $k_{\text{opt}} \approx \frac{\pi}{4}2^{n/2}$, with exact rounding from $\frac{\pi}{4\theta} - \frac{1}{2}$.
**Relevant formulae.** $\sin\theta = 2^{-n/2}$; $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.
**Step-by-step explanation.** $n = 2$: $\theta = \frac{\pi}{6}$, $\frac{\pi}{4\theta} - \frac{1}{2} = 1$ (3). $n = 6$: $\theta \approx 0.1253$, giving $5.77 \to 6$ (4). $n = 8$: $\theta \approx 0.0625$, giving $12.06 \to 12$ (1). $n = 10$: $\theta \approx 0.03126$, giving $24.63 \to 25$ (2). Hence P→3, Q→4, R→1, S→2.
**Why the other options are incorrect.**

**(B)** This exchanges Q and R: the iterations grow with $n$, so $n = 6$ cannot need more than $n = 8$.

**(C)** This exchanges R and S: $n = 8$ gives 12 and $n = 10$ gives 25.

**(D)** This exchanges P and Q: $n = 2$ gives $k = 1$ and $n = 6$ gives $k = 6$.

**Key takeaway.** Adding 2 qubits roughly doubles $k_{\text{opt}}$: $6, 12, 25$.

**Q.56** MATCHING LISTS · Blocks of the circuit

Match each part of the Grover circuit in List-I with its effect in List-II.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | $H^{\otimes n}$ on $\lvert 0\ldots0\rangle$ | **(1)** | returns one bit string with its Born probability |
| **(Q)** | $X$ gates on the 0 bits of $x^*$, multi-controlled $Z$, the same $X$ gates | **(2)** | creates the uniform superposition $\lvert s\rangle$ |
| **(R)** | $H^{\otimes n}X^{\otimes n}$, multi-controlled $Z$, $X^{\otimes n}H^{\otimes n}$ | **(3)** | reflects about $\lvert s\rangle$, up to a global phase |
| **(S)** | measuring all qubits | **(4)** | flips the sign of $\lvert x^*\rangle$ only |

**(A)** ✓ P→2, Q→4, R→3, S→1 **(B)** P→2, Q→4, R→1, S→3

**(C)** P→4, Q→2, R→3, S→1 **(D)** P→2, Q→3, R→4, S→1

**Correct answer: (A)** **Detailed solution**

**Concepts required.** The circuit has four distinct stages: preparation, oracle, diffusion and measurement.

**Relevant formulae.** $\lvert s\rangle = H^{\otimes n}\lvert 0\ldots0\rangle$; $U_\omega = I - 2\lvert x^*\rangle\langle x^*\rvert$; $H^{\otimes n}X^{\otimes n}\,\mathrm{C}^{n-1}\mathrm{Z}\,X^{\otimes n}H^{\otimes n} = -D$.

**Step-by-step explanation.** P: $H^{\otimes n}\lvert 0\ldots0\rangle = \lvert s\rangle$ (2). Q: the oracle recipe of Q.45 flips only $\lvert x^*\rangle$ (4). R: this is the diffusion circuit of Q.47 (3). S: the measurement samples one string from $\lvert c_x\rvert^2$ (1). Hence P→2, Q→4, R→3, S→1.

**Why the other options are incorrect.**

**(B)** This exchanges R and S: the diffusion is unitary and measurement is not.

**(C)** This exchanges P and Q: the Hadamard layer prepares $\lvert s\rangle$, and the $X$-MCZ-$X$ block marks.

**(D)** This exchanges Q and R: the block with $H$ gates is the diffusion, not the oracle.

**Key takeaway.** Oracle: reflect about the marked axis. Diffusion: reflect about $\lvert s\rangle$.

**Q.57** ASSERTION–REASON · Independence of the marked item

**Assertion (A):** The success probability of Grover's algorithm does not depend on which single item is marked.

**Reason (R):** The oracle is a diagonal matrix in the computational basis.

**(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

**(B)** ✓ Both A and R are true, but R is *not* the correct explanation of A

**(C)** A is true, but R is false

**(D)** A is false, but R is true

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Decide A and R first, then ask whether R causes A. The cause here is the symmetry of $\lvert s\rangle$.

**Relevant formulae.** $\langle x^*\lvert s\rangle = 1/\sqrt{N}$ for every basis state $\lvert x^*\rangle$; $U_\omega = \mathrm{diag}\left((-1)^{f(x)}\right)$.

**Step-by-step explanation.** **A is true:** only $\langle x^*\lvert s\rangle = 1/\sqrt{N}$ enters the analysis, and this is the same for every $x^*$. **R is true:** $U_\omega = \mathrm{diag}\left((-1)^{f(x)}\right)$. However, being diagonal does not explain A: an arbitrary diagonal gate need not treat all items alike. The explanation is the symmetry of $\lvert s\rangle$, which has the same overlap with every basis state. Answer: (B).

**Why the other options are incorrect.**

**(A)** R is true but is not the explanation: not every diagonal gate treats all items equally.

**(C)** R is true: $U_\omega$ is diagonal with entries $\pm 1$.

**(D)** A is true: $\langle x^*|s\rangle = 1/\sqrt{N}$ for every $x^*$.

**Key takeaway.** True does not imply explanatory: in Assertion–Reason questions, check for a causal link.

**Q.58** Statement I / II · Oscillation and $\sqrt{N}$ scaling

**Statement I:** In Grover's algorithm, applying more iterations never lowers the success probability.

**Statement II:** For one marked item, the optimal number of iterations grows as $\sqrt{N}$.

**(A)** Both Statement I and Statement II are true

**(B)** Both Statement I and Statement II are false

**(C)** Statement I is true, but Statement II is false

**(D)** ✓ Statement I is false, but Statement II is true

**Correct answer: (D) Detailed solution**

**Concepts required.** $P_k$ oscillates; $k_{\text{opt}} \propto \sqrt{N}$.
**Relevant formulae.** $P_k = \sin^2((2k+1)\theta)$; $k_{\text{opt}} \approx \frac{\pi}{4\theta} \approx \frac{\pi}{4}\sqrt{N}$.
**Step-by-step explanation.** **I is false:** $P_k = \sin^2((2k+1)\theta)$ oscillates. For $N = 4$ we have $P_1 = 1$ but $P_2 = \sin^2 150^\circ = \frac{1}{4}$. **II is true:** $k_{\text{opt}} \approx \frac{\pi}{4\theta} \approx \frac{\pi}{4}\sqrt{N}$. Answer: (D).
**Why the other options are incorrect.**

**(A)** I is false: over-rotation is real, for instance for $N = 4$, $k = 2$.

**(B)** II is true: $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N}$.

**(C)** I is false and II is true, which is the opposite of this option.

**Key takeaway.** One counterexample is enough to reject a "never" statement.

**Paragraph for Questions 59 and 60**

Grover's algorithm runs on $n = 3$ qubits with the single marked item $x^* = |101\rangle$. It starts from the uniform state $|s\rangle$, in which every amplitude is $a = 1/\sqrt{8}$. One iteration is the oracle $U_\omega$ followed by the diffusion $D$. Trace the amplitudes by hand with inversion about the mean.

**Q.59** Paragraph based · Amplitude trace, first iteration

After the first iteration the amplitudes are

**(A)** marked $\frac{3}{2}a$; each unmarked $\frac{1}{2}a$

**(B)** marked $\frac{7}{4}a$; each unmarked $\frac{3}{4}a$

**(C)** ✓ marked $\frac{5}{2}a$; each unmarked $\frac{1}{2}a$

**(D)** marked $\frac{5}{2}a$; each unmarked $\frac{3}{4}a$

**Correct answer: (C) Detailed solution**

**Concepts required.** Inversion about the mean with the mean computed after the oracle.
**Relevant formulae.** $\bar{c} = \frac{1}{N}\sum_x c_x$; marked: $2\bar{c} + a$; unmarked: $2\bar{c} - a$, with $a = 1/\sqrt{8}$ the initial amplitude.
**Step-by-step explanation.** After $U_\omega$: seven entries $a$ and one entry $-a$; the mean is $\frac{7a-a}{8} = \frac{3}{4}a$. Inversion about the mean gives: unmarked $2 \cdot \frac{3}{4}a - a = \frac{1}{2}a$; marked $2 \cdot \frac{3}{4}a + a = \frac{5}{2}a$. *Check:* $(7 \cdot \frac{1}{4} + \frac{25}{4})a^2 = 8a^2 = 1$.
**Why the other options are incorrect.**

**(A)** $\frac{3}{2}a = 2\times$mean omits the term $+a$ for the marked item: $2 \cdot \frac{3}{4}a - (-a) = \frac{5}{2}a$.

**(B)** $\frac{7}{4}a$ and $\frac{3}{4}a$ use the mean as the amplitude and do not preserve the length.

**(D)** The unmarked value is wrong: $\frac{3}{4}a$ is the mean itself.
**Key takeaway.** Marked: $2\text{mean} + a$. Unmarked: $2\text{mean} - a$.

**Q.60** PARAGRAPH BASED · Amplitude trace, second iteration

After the second iteration the probability of measuring $|101\rangle$ is

**(A)** $\frac{25}{32}$ **(B)** $\frac{11}{16}$ **(C)** ✓ $\frac{121}{128}$ **(D)** 1

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Iterate the same two steps; the second mean is smaller because the marked amplitude is now large.
**Relevant formulae.** Repeat the oracle ($c \mapsto -c$ on the marked item) and diffusion ($c_x \mapsto 2\bar{c} - c_x$); $P = |c_{\text{marked}}|^2$; check: $P_2 = \sin^2 5\theta$.
**Step-by-step explanation.** Start from the result of Q.59: unmarked $\frac{1}{2}a$, marked $\frac{5}{2}a$. After $U_\omega$ the marked entry is $-\frac{5}{2}a$, and the mean is $\frac{7\cdot\frac{1}{2}a - \frac{5}{2}a}{8} = \frac{a}{8}$. The marked entry becomes $2\cdot\frac{a}{8} + \frac{5}{2}a = \frac{11}{4}a$, so $P = \frac{121}{16}a^2 = \frac{121}{16\cdot 8} = \frac{121}{128}$. (Each unmarked entry becomes $\frac{a}{4} - \frac{a}{2} = -\frac{a}{4}$.) *Check:* $P_2 = \sin^2 5\theta \approx 0.945$ with $\sin^2\theta = \frac{1}{8}$. A third iteration would over-rotate.
**Why the other options are incorrect.**
**(A)** $\frac{25}{32}$ is the value after the *first* iteration.
**(B)** $\frac{11}{16}$ uses $\frac{11}{4}$ without squaring correctly with $a^2 = \frac{1}{8}$.
**(D)** $P = 1$ would need $5\theta = 90°$ exactly; here it is $103.5°$.
**Key takeaway.** $k = 2$ is optimal for $N = 8$, with $P = \frac{121}{128} \approx 0.945$.

## SECTION D · Advanced Grover and Implementation Q.61–80

**Q.61** Single-correct MCQ · Several solutions: $k_{\text{opt}}$

For $N = 1024$ with $M = 16$ marked items, the optimal number of iterations is

**(A)** 25 **(B)** ✓ 6 **(C)** 5 **(D)** 8

**Correct answer: (B)** **Detailed solution**

**Concepts required.** With $M$ solutions $\sin\theta = \sqrt{M/N}$, and $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N/M}$.
**Relevant formulae.** $\sin\theta = \sqrt{M/N}$; $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right) \approx \frac{\pi}{4}\sqrt{N/M}$.
**Step-by-step explanation.** $\sin\theta = \sqrt{16/1024} = \frac{1}{8}$, so $\theta = 0.12533$ rad. Then $\frac{\pi}{4\theta} - \frac{1}{2} = 6.267 - 0.5 = 5.767$, which rounds to 6. The success probability is $\sin^2 13\theta \approx 0.9966$. (25 is the value for one marked item; $8 = \sqrt{N/M}$ forgets $\frac{\pi}{4}$.)
**Why the other options are incorrect.**
**(A)** 25 is the value for $M = 1$ and $N = 1024$.
**(C)** 5 rounds down instead of to the nearest integer: 5.77 rounds to 6.
**(D)** $8 = \sqrt{N/M}$ forgets $\frac{\pi}{4}$.
**Key takeaway.** More solutions mean fewer iterations: divide $N$ by $M$ before taking the root.

**Q.62** Single-correct MCQ · BBHT threshold for unknown $M$

When $M$ is unknown, the algorithm of Boyer, Brassard, Høyer and Tapp picks $k$ uniformly from $\{0, \ldots, m-1\}$. The average success probability is at least $\frac{1}{4}$ once $m \geq 1/\sin 2\theta$. For $N = 2^{20}$ and $M = 1$ this threshold on $m$ is about

**(A)** ✓ 512 **(B)** 1024 **(C)** 804 **(D)** $2^{20}$

**Correct answer: (A)** **Detailed solution**

**Concepts required.** The BBHT lemma: the average of $P_k$ over $k < m$ is $\frac{1}{2} - \frac{\sin 4m\theta}{4m\sin 2\theta}$.
**Relevant formulae.** $\frac{1}{m}\sum_{k=0}^{m-1}\sin^2((2k+1)\theta) = \frac{1}{2} - \frac{\sin 4m\theta}{4m\sin 2\theta}$, for $k$ uniform on $\{0, \ldots, m-1\}$; for small $\theta$, $\sin 2\theta \approx 2\sin\theta$.
**Step-by-step explanation.** $\sin\theta = 2^{-10}$ is small, so $\sin 2\theta \approx 2\sin\theta = 2^{-9}$ and $\frac{1}{\sin 2\theta} \approx 2^9 = 512$. In general the threshold is about $\frac{1}{2}\sqrt{N/M}$, which comes from the identity $\frac{1}{m}\sum_{k=0}^{m-1}\sin^2((2k+1)\theta) = \frac{1}{2} - \frac{\sin 4m\theta}{4m\sin 2\theta}$; the last term is at most $\frac{1}{4}$ in size once $m \geq 1/\sin 2\theta$.
**Why the other options are incorrect.**
**(B)** $1024 = \sqrt{N}$ forgets $\sin 2\theta \approx 2\sin\theta$.
**(C)** $804 \approx \frac{\pi}{4}\sqrt{N}$ is $k_{\text{opt}}$, not the threshold.
**(D)** $2^{20} = N$ would destroy the speed-up.
**Key takeaway.** Random $k$ below about $\frac{1}{2}\sqrt{N/M}$ already gives average success $\geq \frac{1}{4}$.

**Q.63** Single-correct MCQ · Exact Grover search

To obtain success probability exactly 1, the angle is reduced from $\theta$ to $\theta'$ so that $(2k+1)\theta' = \pi/2$ with $k = \lceil \frac{\pi}{4\theta} - \frac{1}{2} \rceil$. For $N = 16$, $M = 1$ this requires

**(A)** $k = 3$ and $\theta' = \pi/12$

**(B)** $k = 4$ and $\theta' = \pi/18$

**(C)** $k = 2$ and $\theta' = \pi/10$

**(D)** ✓ $k = 3$ and $\theta' = \pi/14$

**Correct answer: (D)** **Detailed solution**

**Concepts required.** An exact search needs $(2k+1)\theta' = \frac{\pi}{2}$ with $\theta' \le \theta$, so round $k$ *up*.
**Relevant formulae.** $k = \lceil \frac{\pi}{4\theta} - \frac{1}{2} \rceil$; $(2k+1)\theta' = \frac{\pi}{2}$ with the reduced angle $\theta' \le \theta$.
**Step-by-step explanation.** $\frac{\pi}{4\theta} - \frac{1}{2} = 2.608$, so the ceiling is $k = 3$. Then $7\theta' = \frac{\pi}{2}$ gives $\theta' = \frac{\pi}{14} \approx 0.2244$, slightly below $\theta \approx 0.2527$ as required. One way to reduce the angle is to mark only with amplitude $\sin\theta'/\sin\theta$ by adding an ancilla rotation. The ceiling matters: rounding down to $k = 2$ would need $\theta' = \frac{\pi}{10} > \theta$, which cannot be reached by reducing the marked weight.
**Why the other options are incorrect.**

**(A)** $\theta' = \frac{\pi}{12}$ gives $7\theta' = \frac{7\pi}{12} \neq \frac{\pi}{2}$.

**(B)** $k = 4$ also satisfies the equation with $\theta' = \frac{\pi}{18}$, but it is not the ceiling and costs one more iteration.

**(C)** $k = 2$ needs $\theta' = \frac{\pi}{10} > \theta$, and the angle cannot be increased by reducing the marked weight.

**Key takeaway.** Round up, then shrink the angle a little, and the probability becomes exactly 1.

**Q.64** Single-correct MCQ · Optimality (BBBV, Zalka)

Which statement about the query complexity of unstructured search is correct?

**(A)** Grover's algorithm is optimal only when $N$ is a power of 2

**(B)** Quantum algorithms need only $O(\log N)$ queries, because $n = \log_2 N$ qubits hold all $N$ items

**(C)** The lower bound $\Omega(\sqrt{N})$ holds only for classical algorithms

**(D)** ✓ Every quantum algorithm needs $\Omega(\sqrt{N})$ oracle queries, so Grover's algorithm is optimal up to a constant

**Correct answer: (D)** **Detailed solution**

**Concepts required.** BBBV lower bound: $\Omega(\sqrt{N})$ queries for black-box search. Zalka: $\frac{\pi}{4}\sqrt{N}$ is exactly optimal.
**Relevant formulae.** Quantum query lower bound: $\Omega(\sqrt{N/M})$; Grover uses $\frac{\pi}{4}\sqrt{N/M}$; classical cost: $\Theta(N/M)$.
**Step-by-step explanation.** The lower bound of Bennett, Bernstein, Brassard and Vazirani states that any quantum algorithm with black-box access needs $\Omega(\sqrt{N})$ queries; Zalka showed that $\frac{\pi}{4}\sqrt{N}$ is optimal even including the constant. Superposition of $N$ items does not give $O(\log N)$ search, since a measurement returns only one string and the marked amplitude grows slowly.
**Why the other options are incorrect.**

**(A)** The result does not depend on $N$ being a power of 2.

**(B)** Holding $N$ amplitudes in $n$ qubits does not allow reading them out, and $\log N$ queries would violate the lower bound.

**(C)** Classical search needs $\Theta(N)$; the bound $\Omega(\sqrt{N})$ is the quantum one.

**Key takeaway.** No black-box quantum search can beat $\sqrt{N}$ by more than a constant.

**Q.65** Single-correct MCQ · Reading gate statistics

The notebook's full-circuit statistics for $n = 15$ and $k = 142$ report 4275 `h` gates, 5964 `x` gates and 284 multi-controlled $Z$ gates. In one iteration the diffusion uses $2n$ `x` gates and the oracle uses two `x` gates for each 0 in $x^*$. The number of 0s in the marked string is

**(A)** 9 **(B)** 21 **(C)** ✓ 6 **(D)** 12

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Per iteration the circuit has $2n$ H, $2n + 2z$ X and two multi-controlled $Z$ gates; the first H layer adds $n$ H gates.

**Relevant formulae.** `h` count $= n + 2nk$; `x` count $= k(2n + 2z)$; multi-controlled $Z$ count $= 2k$, with $k$ iterations and $z$ zeros in $x^*$.

**Step-by-step explanation.** `x` gates per iteration: $\frac{5964}{142} = 42$. Of these, $2n = 30$ belong to the diffusion, which leaves $12 = 2z$ for the oracle, so $z = 6$. The marked string in that run, 011001010011111, indeed has six 0s. *Check of the other counts:* `h`: $15 + 142 \times 30 = 4275$; multi-controlled $Z$: $2 \times 142 = 284$.

**Why the other options are incorrect.**

**(A)** 9 is the number of 1s in the marked string, not of 0s.

**(B)** 21 divides 42 by 2 without first removing the $2n$ gates of the diffusion.

**(D)** $12 = 2z$ counts the oracle's `x` gates, not the zeros.

**Key takeaway.** Gate statistics can be decoded with the formula $4n + 2z + 2$: here $60 + 12 + 2 = 74$ gates per iteration.

**Q.66** Single-correct MCQ · Amplitude amplification cost

Amplitude amplification boosts a procedure $\mathcal{A}$ that succeeds with probability $a = 0.01$ ($\theta \approx 0.10017$). At the optimal $k$, one uses $\mathcal{A}$ or $\mathcal{A}^{-1}$ in total

**(A)** 7 times **(B)** ✓ 15 times **(C)** 100 times **(D)** 8 times

**Correct answer: (B)** **Detailed solution**

**Concepts required.** $Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$: each iteration uses $\mathcal{A}$ and $\mathcal{A}^{-1}$ once, plus one initial $\mathcal{A}$.

**Relevant formulae.** $\sin^2\theta = a$, the success probability of $\mathcal{A}$; $Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$; total uses of $\mathcal{A}$ and $\mathcal{A}^{-1}$: $2k + 1$.

**Step-by-step explanation.** Optimal $k$: $\frac{\pi}{4\theta} - \frac{1}{2} = 7.34$, so $k = 7$. The initial state uses $\mathcal{A}$ once, and each iteration $Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$ uses $\mathcal{A}$ and $\mathcal{A}^{-1}$ once each. Total $2k + 1 = 15$ uses. Classical repetition needs about $1/a = 100$ runs on average.

**Why the other options are incorrect.**

**(A)** 7 is the iteration count, not the number of uses of $\mathcal{A}$.

**(C)** 100 is the classical repetition count.

**(D)** 8 is $k + 1$ and ignores the two uses per iteration.

**Key takeaway.** $2k + 1$ uses of $\mathcal{A}$, compared with $1/a$ classical repetitions.

**Q.67** Single-correct MCQ · Notebook memory estimate

On the CPU state-vector backend the notebook estimates the memory as $2^n \times 16$ bytes for the simulator's state plus one equal host copy, because `RUN_ITERATION_SWEEP = True` holds two states at once. Its report for $n = 15$ is 1.05 MB. With the same settings and $n = 28$ the estimate is

**(A)** 2 GiB **(B)** 16 GiB **(C)** 4 GiB **(D)** ✓ 8 GiB

**Correct answer: (D)** **Detailed solution**

**Concepts required.** State vector: $2^n$ amplitudes of 16 bytes in double precision. The sweep keeps one additional copy.

**Relevant formulae.** Memory $= (1 + c) \times 2^n \times 16$ bytes, with $c = 1$ extra copy of the state; 1 GiB $= 2^{30}$ bytes.

**Step-by-step explanation.** One state: $2^{28} \times 16 = 2^{32}$ bytes $= 4$ GiB. Two states: 8 GiB. *Check against the report:* for $n = 15$, $2 \times 2^{15} \times 16 = 1\,048\,576$ bytes $= 1.05$ MB in decimal units. Each extra qubit doubles the requirement. Switching off the sweep removes the copy and halves the figure.

**Why the other options are incorrect.**

**(A)** 2 GiB is one state at $n = 27$.

**(B)** 16 GiB is the value for $n = 29$.

**(C)** 4 GiB is one state, without the extra copy.

**Key takeaway.** Memory doubles per qubit; settings such as the sweep can double it again.

**Q.68** Single-correct MCQ · Key search

A cipher with a 128-bit key is attacked by Grover's algorithm (one correct key, cost measured in iterations). The number of iterations is about

**(A)** ✓ $\frac{\pi}{4} \cdot 2^{64}$ **(B)** $\frac{\pi}{4} \cdot 2^{127}$ **(C)** $2^{127}$ **(D)** 128

**Correct answer: (A)** **Detailed solution**

**Concepts required.** $k \approx \frac{\pi}{4}\sqrt{N}$ with $N = 2^{128}$. Symmetric keys lose half their bits against Grover.

**Relevant formulae.** $k \approx \frac{\pi}{4}\sqrt{N}$, with $N = 2^b$ keys of $b$ bits.

**Step-by-step explanation.** $N = 2^{128}$, so $k \approx \frac{\pi}{4}\sqrt{N} = \frac{\pi}{4} \cdot 2^{64}$. The classical average is $2^{127}$. Doubling the key length restores the original margin. Each iteration also has to run the cipher reversibly, so the real cost is much larger than the iteration count.

**Why the other options are incorrect.**

**(B)** This uses $\sqrt{N}$ with the wrong exponent.

**(C)** $2^{127}$ is the classical average.

**(D)** 128 is the key length, and search is not polynomial in it.

**Key takeaway.** Doubling the key length cancels Grover's advantage.

**Q.69** Single-correct MCQ · Noise and depth

A simple noise model multiplies the signal by $f$ per iteration: $P_k \approx f^k \sin^2((2k+1)\theta)+(1-f^k)/N$. For $n = 10$, $k = 25$, $f = 0.99$ (take $0.99^{25} \approx 0.78$), the success probability is about

**(A)** 0.99 **(B)** 0.50 **(C)** ✓ 0.78 **(D)** 0.25

**Correct answer: (C)** **Detailed solution**

**Concepts required.** The signal decays as $f^k$, while the background $(1 - f^k)/N$ is very small.
**Relevant formulae.** $P_k \approx f^k \sin^2((2k+1)\theta)+(1-f^k)/N$, with $f$ the signal retained per iteration.
**Step-by-step explanation.** For $n = 10$, $k = 25$ is optimal and $\sin^2(51\theta) \approx 0.9995$, while $(1 - f^k)/N \approx 0.22/1024$ is negligible. So $P \approx 0.78 \times 0.9995 \approx 0.78$. Under this model, a depth of $k$ iterations costs a factor $f^k$; since $k \propto \sqrt{N}$, this decay is what limits large searches on noisy hardware.
**Why the other options are incorrect.**

**(A)** 0.99 is the fidelity of one iteration, not of 25.

**(B)** 0.50 has no basis in the model.

**(D)** 0.25 confuses $k = 25$ with a probability.

**Key takeaway.** Long Grover circuits suffer from noise, since $k \propto \sqrt{N}$.

**Q.70** Single-correct MCQ · Toffoli count

A multi-controlled $Z$ with $m$ controls can be built with $m-1$ clean ancillas, $2(m-1)$ Toffoli gates and one CZ. For the $n = 10$ Grover register ($m = 9$), one full iteration (oracle and diffusion) needs how many Toffoli gates?

**(A)** ✓ 32 **(B)** 36 **(C)** 16 **(D)** 18

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Each iteration has two multi-controlled $Z$ gates. Each costs $2(m-1)$ Toffolis with this construction.
**Relevant formulae.** Multi-controlled $Z$ with $m$ controls: $m - 1$ clean ancillas, $2(m - 1)$ Toffoli gates, one CZ. Two such gates per iteration.
**Step-by-step explanation.** One multi-controlled $Z$ with $m = 9$: $2(9 - 1) = 16$ Toffolis. An iteration has two of them, one in the oracle and one in the diffusion, so 32 Toffolis. The gate count per iteration is linear in $n$.
**Why the other options are incorrect.**

**(B)** $36 = 4m$ also uses $m$ instead of $m - 1$.

**(C)** 16 counts only one of the two multi-controlled $Z$ gates.

**(D)** $18 = 2m$ uses the wrong count per gate.

**Key takeaway.** The cost is linear in $n$ per iteration, and most of it is in the multi-controlled gates.

**Q.71** Single-correct MCQ · Simulation cost

On a state-vector simulator the work grows as (iterations) × (amplitudes per gate) $\sim 2^{n/2} \cdot 2^n$. Adding 2 qubits multiplies the total work by about

**(A)** 16 **(B)** ✓ 8 **(C)** 4 **(D)** $2\sqrt{2}$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Work $\sim 2^{1.5n}$: amplitudes double per qubit and iterations grow by $\sqrt{2}$ per qubit.
**Relevant formulae.** Work $\sim k \cdot 2^n \sim 2^{n/2} \cdot 2^n = 2^{1.5n}$.
**Step-by-step explanation.** Work $\propto 2^{1.5n}$, so $n \to n+2$ gives $2^{1.5\times 2} = 2^3 = 8$: amplitudes grow by 4 and iterations by 2. The number of gates per iteration also grows slowly with $n$, which this estimate ignores.
**Why the other options are incorrect.**
**(A)** 16 double-counts: it would need work $\sim 2^{2n}$.
**(C)** 4 counts only the growth in amplitudes.
**(D)** $2\sqrt{2}$ is the factor for adding *one* qubit.
**Key takeaway.** The classical simulation cost is about $N^{1.5}$.

**Q.72** Single-correct MCQ · Reading the sweep code

The notebook's sweep computes

```
sweep_max = 2 * k_optimal + 1
stride = max(1, ceil(sweep_max / SWEEP_MAX_POINTS))
```

It records the point $k = 0$ and then evolves by `stride` iterations `while k_done + stride <= sweep_max`, recording one point each time. For $k_{\text{opt}} = 142$ and `SWEEP_MAX_POINTS = 60`, the number of recorded points is

**(A)** 57 **(B)** 285 **(C)** ✓ 58 **(D)** 60

**Correct answer: (C)** **Detailed solution**

**Concepts required.** Read the loop: the point $k = 0$ is recorded first and then one point per stride while $k$ stays within the range.
**Relevant formulae.** `stride` $= \lceil$`sweep_max/SWEEP_MAX_POINTS`$\rceil$; points $= 1 + \lfloor$`sweep_max/stride`$\rfloor$.
**Step-by-step explanation.** `sweep_max` $= 285$, and `stride` $= \lceil 285/60 \rceil = \lceil 4.75 \rceil = 5$. The loop records $k = 5, 10, \ldots, 285$, which is $\frac{285}{5} = 57$ points, and with $k = 0$ the total is 58. The stored output of the notebook says "58 points". The rounding up means the loop may record fewer points than the requested maximum of 60, never more.
**Why the other options are incorrect.**
**(A)** 57 forgets the point $k = 0$.
**(B)** 285 is the range of $k$, which would be recorded only with stride 1.
**(D)** 60 is the requested maximum; rounding the stride up gives fewer.
**Key takeaway.** `SWEEP_MAX_POINTS` is an upper limit: compute the stride to find the real number of points.

**Q.73** Single-correct MCQ · Interpreting 2000/2000 shots

A notebook run has theoretical success probability $P = 0.99998683$ and records 2000 marked counts out of 2000 shots. For independent shots, the probability of seeing all 2000 shots succeed is about

**(A)** 0.99999 **(B)** 0.026 **(C)** 0.500 **(D)** ✓ 0.974

**Correct answer: (D) Detailed solution**

**Concepts required.** Independent shots multiply: $P(\text{all } S \text{ succeed}) = P^S \approx e^{-S(1-P)}$ for $P$ near 1.
**Relevant formulae.** $P(\text{all } S \text{ shots succeed}) = P^S$; $\ln P \approx -(1-P)$ for $P$ near 1.
**Step-by-step explanation.** All shots succeed with probability $P^{2000}$. Using $\ln P \approx -(1-P) = -1.317 \times 10^{-5}$:

$$P^{2000} \approx e^{-2000 \times 1.317 \times 10^{-5}} = e^{-0.0263} \approx 0.974.$$

So 2000/2000 is the expected outcome. It does *not* show that $P$ is exactly 1: the run is consistent with a small failure probability that 2000 shots cannot resolve.
**Why the other options are incorrect.**
**(A)** 0.99999 is $P$ itself, the value for a single shot.
**(B)** 0.026 is about the probability of at least one failure, the complement of the answer.
**(C)** 0.500 has no basis: $2000(1-P)$ is much smaller than $\ln 2$.
**Key takeaway.** A perfect count is consistent with $P$ slightly below 1; finite shots cannot prove $P = 1$.

**Q.74** Single-correct MCQ · Repeat-until-success cost

Let $N = 8$, $M = 1$, and repeat a run with a fixed $k$ until the marked item is found, counting only oracle queries in the Grover iterations. Then $P_1 = \frac{25}{32}$ and $P_2 = \frac{121}{128}$. The expected numbers of queries for $k = 1$ and $k = 2$ are about

**(A)** 1 and 2 **(B)** 2.12 and 1.28 **(C)** ✓ 1.28 and 2.12 **(D)** 0.78 and 1.89

**Correct answer: (C) Detailed solution**

**Concepts required.** Expected cost of repeat-until-success is $k/P_k$.
**Relevant formulae.** Number of runs is geometric with mean $1/P_k$; expected queries $= k/P_k$, with $k$ queries per run.
**Step-by-step explanation.** The number of runs is geometric with mean $1/P_k$, and each run costs $k$ queries, so the expected cost is $k/P_k$. $k = 1$: $\frac{32}{25} = 1.28$. $k = 2$: $2 \times \frac{128}{121} \approx 2.12$. So $k = 1$ is cheaper here, although $k = 2$ is the value that maximises the success probability of a single run: the two criteria are different.
**Why the other options are incorrect.**
**(A)** Those are $k$, which ignores repetition after failed runs.
**(B)** The values are in the wrong order.
**(D)** These are $P_1$ and $kP_2$ rather than $k/P_k$.
**Key takeaway.** The $k$ that maximises $P_k$ need not minimise the expected cost.

**Q.75** MATCHING LISTS · Query complexities

Match each task in List-I with its query complexity in List-II.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | Unstructured search, classical | **(1)** | $\Theta(N^{1/3})$ |
| **(Q)** | Unstructured search, quantum | **(2)** | $\Theta(N^{2/3})$ |
| **(R)** | Collision finding in a 2-to-1 function, quantum | **(3)** | $\Theta(N)$ |
| **(S)** | Element distinctness, quantum | **(4)** | $\Theta(N^{1/2})$ |

**(A)** ✓ P→3, Q→4, R→1, S→2 **(B)** P→4, Q→3, R→1, S→2

**(C)** P→3, Q→1, R→4, S→2 **(D)** P→3, Q→4, R→2, S→1

**Correct answer: (A)** **Detailed solution**

**Concepts required.** Known query complexities: search $N$ vs. $\sqrt{N}$, collision $N^{1/3}$, element distinctness $N^{2/3}$.

**Relevant formulae.** Queries as $N$ grows: $\Theta(N)$, $\Theta(N^{1/2})$, $\Theta(N^{1/3})$, $\Theta(N^{2/3})$, where $\Theta$ means growth up to constant factors.

**Step-by-step explanation.** Classical search must examine a constant fraction of the items: $\Theta(N)$ (3). Grover: $\Theta(N^{1/2})$ (4). Brassard–Høyer–Tapp collision finding, optimal by Aaronson–Shi: $\Theta(N^{1/3})$ (1). Ambainis's quantum walk for element distinctness: $\Theta(N^{2/3})$ (2). Hence P→3, Q→4, R→1, S→2.

**Why the other options are incorrect.**

**(B)** This exchanges classical and quantum search.

**(C)** This assigns $N^{1/3}$ to unstructured search, which contradicts the lower bound.

**(D)** This exchanges collision and element distinctness; the promise of 2-to-1 makes collision easier.

**Key takeaway.** More structure leads to lower query complexity.

**Q.76** MATCHING LISTS · Variants of Grover

Match each method in List-I with its main idea in List-II.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | Amplitude amplification | **(1)** | uses randomly chosen $k$ from a growing range when $M$ is unknown |
| **(Q)** | Quantum counting | **(2)** | changes the reflection phases so that extra steps cannot over-rotate |
| **(R)** | BBHT search | **(3)** | replaces $H^{\otimes n}$ by any state-preparation procedure $\mathcal{A}$ |
| **(S)** | Fixed-point search | **(4)** | applies phase estimation to $G$ to find $M$ |

**(A)** P→3, Q→1, R→4, S→2 **(B)** ✓ P→3, Q→4, R→1, S→2

**(C)** P→3, Q→4, R→2, S→1 **(D)** P→4, Q→3, R→1, S→2

**Correct answer: (B)** **Detailed solution**

**Concepts required.** Each variant answers one weakness: arbitrary start, unknown $M$, or over-rotation.

**Relevant formulae.** $G$ has eigenvalues $e^{\pm 2i\theta}$; $Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$; $\sin^2\theta = M/N$.

**Step-by-step explanation.** Amplitude amplification generalises $|s\rangle$ to $\mathcal{A}|0\rangle$ (3). Quantum counting reads $\theta$ from the eigenvalues $e^{\pm 2i\theta}$ of $G$, which yields $M$ (4). BBHT draws $k$ at random from a range that grows geometrically (1). Fixed-point search uses reflections with phase $\pi/3$ and similar choices rather than $\pi$ (2). Hence P→3, Q→4, R→1, S→2.

**Why the other options are incorrect.**

**(A)** This exchanges Q and R.

**(C)** This exchanges R and S; fixed-point search changes the phases and BBHT randomises $k$.

**(D)** This exchanges P and Q; quantum counting uses phase estimation.

**Key takeaway.** Learn the problem each variant solves, not only its name.

**Q.77** Assertion–Reason · Grover and NP-completeness

**Assertion (A):** Grover's algorithm solves NP-complete problems in polynomial time.

**Reason (R):** For a search over $2^n$ candidates, Grover's algorithm uses about $\frac{\pi}{4}2^{n/2}$ iterations.

**(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

**(B)** Both A and R are true, but R is *not* the correct explanation of A

**(C)** A is true, but R is false

**(D)** ✓ A is false, but R is true

**Correct answer: (D) Detailed solution**

**Concepts required.** Quadratic speed-up does not make an exponential polynomial.

**Relevant formulae.** $k \approx \frac{\pi}{4}\sqrt{N} = \frac{\pi}{4}2^{n/2}$, with $N = 2^n$ candidates.

**Step-by-step explanation.** **R is true:** this is $\frac{\pi}{4}\sqrt{N}$ with $N = 2^n$. **A is false:** $2^{n/2}$ is still exponential in $n$. The speed-up is quadratic, and the black-box lower bound shows that generic search cannot do better. Answer: (D).

**Why the other options are incorrect.**

**(A)** A is false, so this option fails.

**(B)** A is false, so this option fails even though R is true.

**(C)** A is false and R is true, the opposite of this option.

**Key takeaway.** $2^{n/2}$ is still exponential.

**Q.78** Statement I / II · Unknown $M$; flat curve at $M = N/2$

**Statement I:** The BBHT algorithm finds a marked item with an expected $O(\sqrt{N/M})$ queries without knowing $M$ in advance.

**Statement II:** If exactly half of the items are marked, ordinary Grover iterations leave the success probability at $\frac{1}{2}$.

**(A)** ✓ Both Statement I and Statement II are true

**(B)** Both Statement I and Statement II are false

**(C)** Statement I is true, but Statement II is false

**(D)** Statement I is false, but Statement II is true

**Correct answer: (A) Detailed solution**

**Concepts required.** BBHT handles unknown $M$; and $\theta = 45°$ gives a flat success curve.

**Relevant formulae.** Expected BBHT queries: $O(\sqrt{N/M})$; $\sin^2((2k+1)45°) = \frac{1}{2}$ for every integer $k$.

**Step-by-step explanation.** **I is true:** a growing range for the random $k$ reaches $m \sim \sqrt{N/M}$

after a geometric series of attempts, whose total cost is dominated by the final terms. **II is true:** $\theta = 45°$ and $\sin^2((2k+1)45°) = \frac{1}{2}$ for all $k$ (see Q.50). Answer: (A).

**Why the other options are incorrect.**

**(B)** Both statements are true, not false.

**(C)** II is also true, as $\sin^2$ of odd multiples of $45°$ is $\frac{1}{2}$.

**(D)** I is also true: the expected cost is $O(\sqrt{N/M})$.

**Key takeaway.** Unknown $M$ is solved, and $M = N/2$ is not a problem since simply measuring works.

## Paragraph for Questions 79 and 80

**Quantum counting.** A search space of $N = 64$ items contains an unknown number $M$ of marked items; in fact $M = 16$. The Grover operator $G$ has eigenvalues $e^{\pm 2i\theta}$ on the plane containing $|s\rangle$, with $\sin^2\theta = M/N$. Phase estimation of $G$ with $t = 3$ counting qubits is run on $|s\rangle$, which has equal weight on the two eigenvectors. The outcome $y \in \{0, \ldots, 7\}$ gives the phase estimate $\varphi \approx y/8$, where $e^{2\pi i\varphi}$ is the eigenvalue. From this one sets $\tilde{\theta} = \pi y/8$ and $\tilde{M} = N\sin^2\tilde{\theta}$.

**Q.79** Paragraph based · Quantum counting outcomes

The two most likely outcomes are

**(A)** $y = 2$ and $y = 6$ **(B)** ✓ $y = 1$ and $y = 7$ **(C)** $y = 1$ and $y = 2$ **(D)** $y = 0$ and $y = 4$

**Correct answer: (B)** **Detailed solution**

**Concepts required.** $\varphi = \theta/\pi$ and $1 - \theta/\pi$. QPE peaks at the grid point nearest to $2^t\varphi$.

**Relevant formulae.** $e^{2\pi i\varphi} = e^{\pm 2i\theta}$ gives $\varphi = \frac{\theta}{\pi}$ and $1 - \frac{\theta}{\pi}$; phase estimation peaks at the integer $y$ nearest to $2^t\varphi$, with $t$ counting qubits.

**Step-by-step explanation.** $\theta = \arcsin\sqrt{16/64} = \arcsin\frac{1}{2} = \frac{\pi}{6}$, so the eigenphases are $\varphi = \frac{\theta}{\pi} = \frac{1}{6}$ and $\frac{5}{6}$. On the grid $y/8$ these lie at $y = \frac{8}{6} = 1.33$ and $y = \frac{40}{6} = 6.67$; the closest integers are 1 and 7. Each has probability about 0.35. (The next most likely are $y = 2$ and $y = 6$, each about 0.09.)

**Why the other options are incorrect.**

**(A)** $y = 2, 6$ are the next most likely, each with about 0.09.

**(C)** Both eigenphases contribute, and they are symmetric: $y$ and $8 - y$.

**(D)** $y = 0, 4$ would be $\theta = 0$ and $\theta = \frac{\pi}{2}$.

**Key takeaway.** Outcomes come in pairs $y$ and $2^t - y$, which give the same $\sin^2\tilde{\theta}$.

**Q.80** Paragraph based · Quantum counting estimate

If $y = 1$ is obtained, the estimate of $M$ is

**(A)** $\tilde{M} = 32$ **(B)** $\tilde{M} = 8$ **(C)** ✓ $\tilde{M} \approx 9.4$ **(D)** $\tilde{M} = 16$

**Correct answer: (C)** **Detailed solution**

**Concepts required.** $\tilde{M} = N\sin^2(\pi y/2^t)$; $\sin^2\frac{\pi}{8} = \frac{1-\cos(\pi/4)}{2}$.

**Relevant formulae.** $\tilde{\theta} = \frac{\pi y}{2^t}$; $\tilde{M} = N\sin^2\tilde{\theta}$; $\sin^2\alpha = \frac{1-\cos 2\alpha}{2}$.

**Step-by-step explanation.** $\tilde{\theta} = \frac{\pi}{8}$ and $\tilde{M} = 64\sin^2\frac{\pi}{8} = 64 \cdot \frac{1-\cos(\pi/4)}{2} = 32(1 - \frac{\sqrt{2}}{2}) \approx 9.37$. The true value 16 lies between the estimates from $y = 1$ (9.4) and $y = 2$ (32): three counting qubits are too few for an accurate count. Each additional counting qubit halves the error in $\theta$.

**Why the other options are incorrect.**

**(A)** 32 is the estimate from $y = 2$.

**(B)** 8 uses $\sin^2 \frac{\pi}{8} \approx \frac{1}{8}$ and is not the correct value.

**(D)** 16 is the true value, but $y = 1$ gives a coarse estimate.

**Key takeaway.** More counting qubits give a finer grid and a better estimate of $M$.